

ORGANIZE

The Functional Medicine Matrix



ROBERT LUBY, MD

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Disclosures

Robert Luby, MD, disclosed no relevant financial relationships with any ineligible companies.

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Learning Objectives

At the conclusion of this segment, successful participants will be able to:

1. Identify the sections and features of the IFM Functional Medicine Matrix.
2. Explain how the IFM Matrix is used to organize the patient's data and enhance the patient evaluation process.
3. Explain the physiological dysfunctions and interrelationships among the seven domains of clinical imbalance.

IFM Clinical Skills and Performance Objectives

THIS SEGMENT WILL ENABLE PARTICIPANTS TO DEVELOP & APPLY THESE SKILLS IN THEIR PRACTICE:

1. Populate the data gathered during the patient evaluation onto the IFM Matrix.
2. Utilize the IFM Matrix to identify root causes and generate an explanation of the most important physiological dysfunctions and clinical imbalances contributing to the patient's condition.

GOTOIT Functional Medicine Framework



GATHER oneself and clinical findings on the IFM Functional Medicine Timeline.



ORGANIZE clinical findings and imbalances on the IFM Functional Medicine Matrix.



TELL the patient's story to ensure a shared understanding.



ORDER and prioritize interventions feasible for the patient.



INITIATE referrals, evaluative tests, and therapeutic modalities.



TRACK and adjust evaluative and therapeutic regimens.

Long Description: GO TO IT Framework

- Go to it functional medicine framework
- G
- Gather oneself and clinical findings on the I F M functional medicine timeline
- O
- Organize clinical findings and imbalances on the I F M functional medicine matrix
- T
- Tell the patient's story to ensure a shared understanding
- O
- Order and prioritize interventions feasible for the patient
- I
- Initiate referrals, evaluative tests, and therapeutic modalities
- T
- Track and adjust evaluate and therapeutic regimens

GOTOIT Functional Medicine Framework



GATHER oneself and clinical findings on the IFM Functional Medicine Timeline.



ORGANIZE clinical findings and imbalances on the IFM Functional Medicine Matrix.



TELL the patient's story to ensure a shared understanding.



ORDER and prioritize interventions feasible for the patient.



INITIATE referrals, evaluative tests, and therapeutic modalities.



TRACK and adjust evaluative and therapeutic regimens.

Clinical Focus

- Description and purpose of the Organize step of the GOTOIT method.
- Role of the IFM Matrix in transforming and improving the evaluation of the patient.



Organize

Organize the patient's data gathered in the previous step and document them on the IFM Timeline and Matrix to identify the personalized root causes of current concerns, dysfunctions, and conditions and to begin to prioritize the order (sequence) of interventions.

The ORGANIZE Step of GOTOIT

Populate the MATRIX using the information gathered on the TIMELINE:

- Antecedents, triggers, and mediators.
- Modifiable lifestyle factors.
- Mental, emotional, and spiritual factors.
- Clinical findings and imbalances.



GOTOIT

The sequence of steps used in the practice of functional medicine:

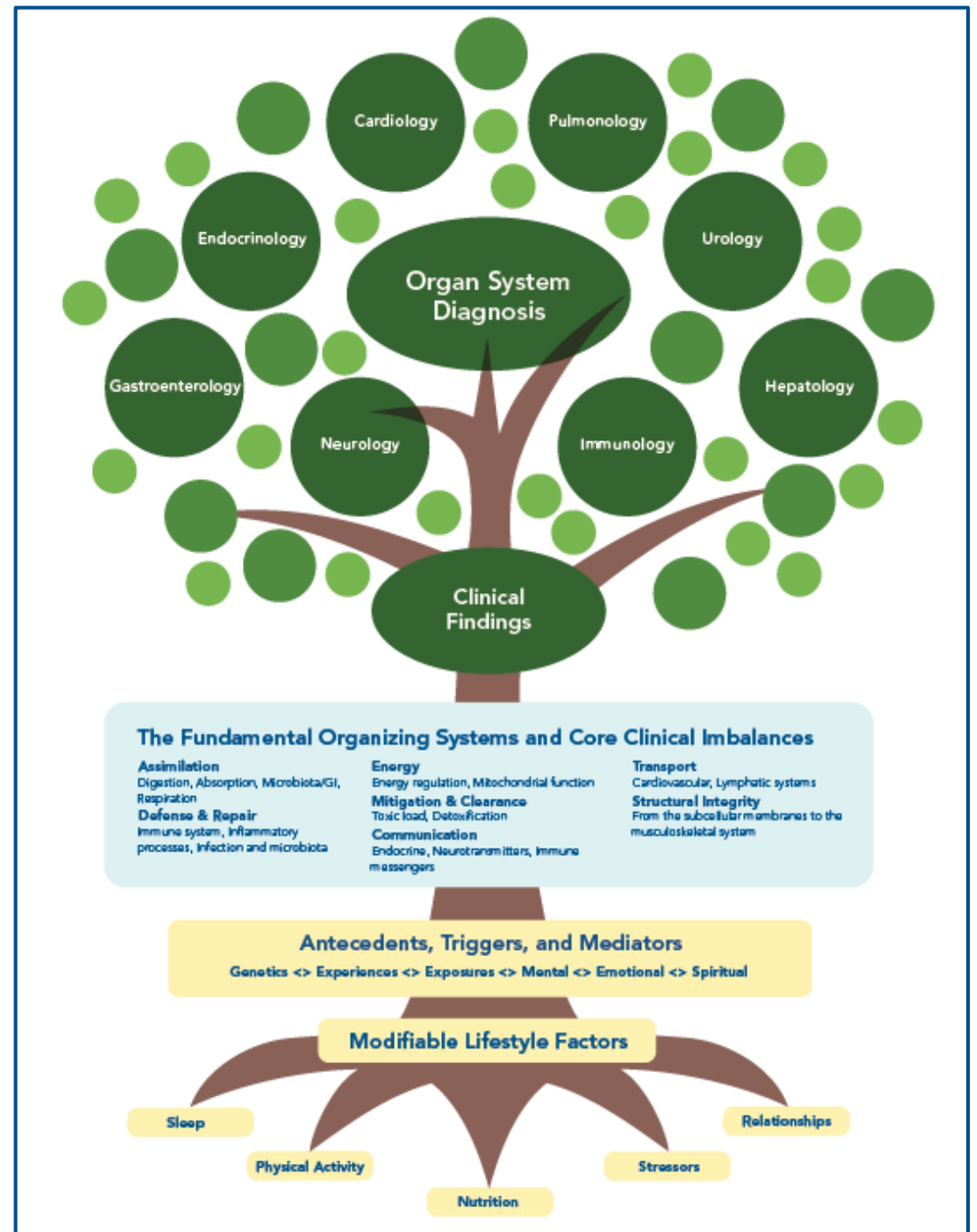
GOTOIT Step and Definition	IFM Tools (examples)	Clinical Actions
G – GATHER oneself with a brief personal practice prior to prepare for each patient encounter. Take a detailed functional medicine history that includes the essential health events and features of a patient's life documented on the IFM Timeline in the process of organizing the patient's data.	• Cultivating Self Awareness and Mindfulness • Adult Intake Form • Medical Symptom Questionnaire (MSQ) • PROMIS Questionnaires • Diet, Nutrition, and Lifestyle Journal • Functional Medicine Tree • Functional Medicine Timeline	• Use the Cultivating Self-Awareness and Mindfulness document to help gather yourself before every patient encounter. • The Adult Intake Form is used to gather initial intake information. Whereas, the MSQ, PROMIS Questionnaires, and Diet, Nutrition, and Lifestyle Journal help gather initial symptom, function, and dietary information. • Map root causes (antecedents, triggers, mediators (ATMs)) and effects (clinical findings, current concerns) of physiological dysfunction on the Functional Medicine Timeline.
O – ORGANIZE the data gathered in the previous step and document them on the IFM Timeline and Matrix to identify the personalized root causes of current concerns, dysfunctions, and conditions, and to begin to prioritize the order (sequence) of interventions.	• Functional Medicine Tree • Functional Medicine Timeline • Functional Medicine Matrix • Adult Intake Form • Medical Symptom Questionnaire (MSQ) • PROMIS Questionnaires • Diet, Nutrition, and Lifestyle Journal	• Using the Functional Medicine Matrix, populate ATMs, clinical findings, and current concerns from the Timeline. Add modifiable lifestyle factors (MLFs) and mental, emotional, spiritual factors (MES) to support the identification of the most important areas of physiological dysfunction.
T – TELL the life story back to the patient to ensure accuracy and a shared understanding. This dialogue, focused on the Timeline, highlights how factors elicited in the Gather step may have contributed to current or future clinical findings, concerns, health status and trajectory. The patient is asked to join in correcting the story, engendering a context of true partnership.	• Functional Medicine Matrix • Functional Medicine Timeline • Functional Medicine Tree	• The Functional Medicine Timeline is used to re-tell the patient's story while correlating prior events and root causes with current concerns or conditions. Ask the patient to join in for corrections or additional context. • Use the Functional Medicine Tree to explain how the root causes have contributed to the patient's condition.

Long Description: The Organize Step of GO TO IT

- Clinical Toolkit item: G O T O I T, Steps Practicing Functional Medicine, accessible version available in course materials.

Organizing the Data Nodes of the Matrix:

- Distinguishing features that inform the Functional Medicine Patient Evaluation



Long Description: Organizing the Data

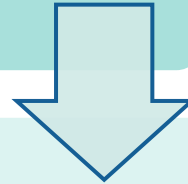
- The practitioner can organize the distinguishing features of the functional medicine patient evaluation like the structure of a tree. The roots of the tree represent the modifiable lifestyle factors including sleep, physical activity, nutrition, stressors, and relationships. Moving upward, the base of the trunk represents antecedents, triggers, and mediators including genetics, epigenetics, exposures, mental health, emotional health, and spiritual health. The middle of the trunk is the fundamental organizing systems and core clinical imbalances. These are: assimilation, which includes digestion, absorption, microbiota or G I, and respiration; defense and repair, which includes the immune system, inflammatory processes, and infection and microbiota; energy, which includes energy regulation and mitochondrial function; mitigation and clearance, which includes toxic load and detoxification; communication, which includes endocrine, neurotransmitters, and immune messengers; transport, which includes cardiovascular and lymphatic systems; and structural integrity from the subcellular membranes to the musculoskeletal system. The top of the trunk represents clinical findings, and the top branches are the organ system diagnosis. The leaves represent the organ systems including neurology, gastroenterology, endocrinology, cardiology, pulmonology, urology, hepatology, and immunology.

20th Century Medicine

Identify the symptoms
and the effects

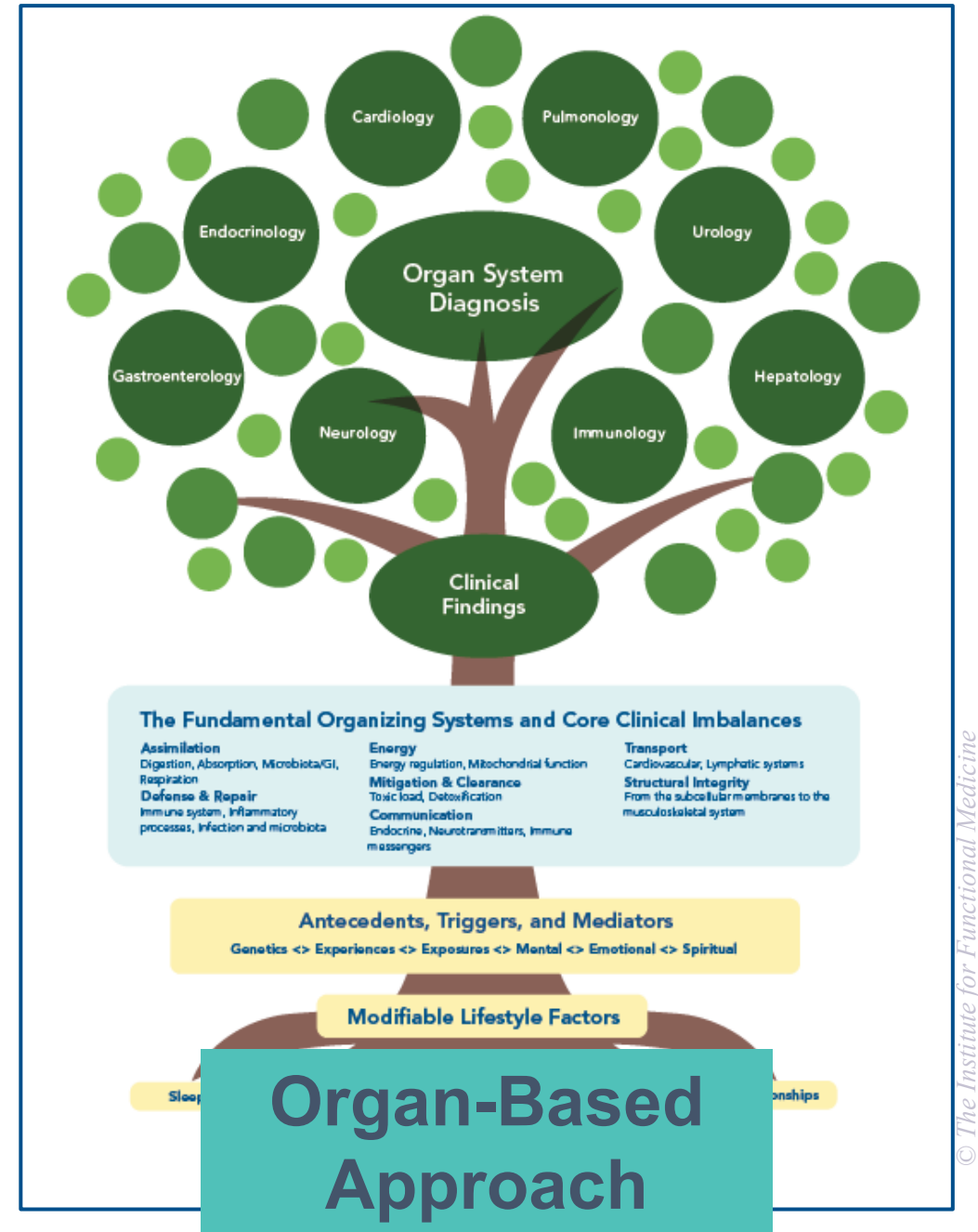


Diagnose the disease



Treat the symptoms
and the effects

Accessible version in your toolkit.



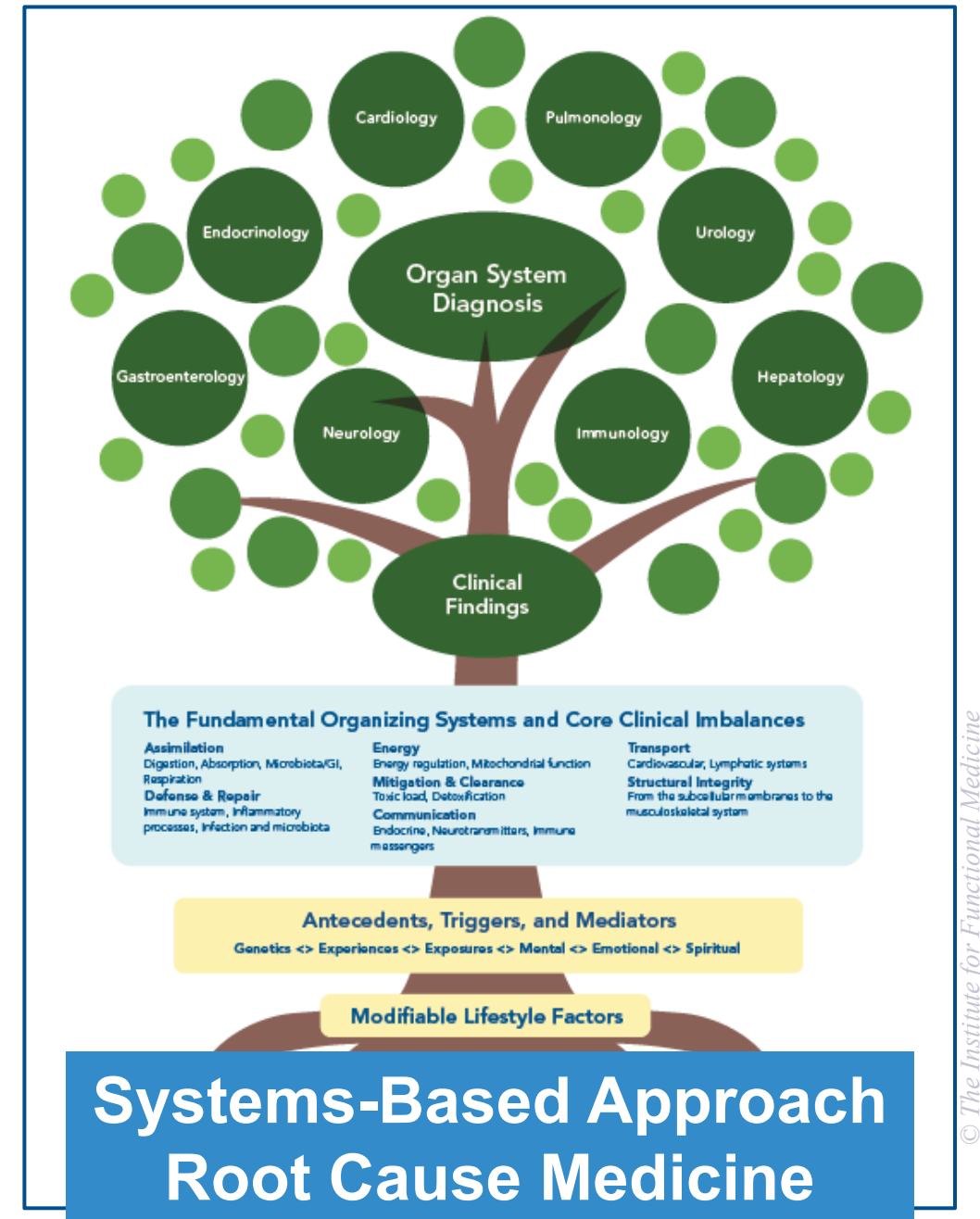
21st Century Medicine

Identify the personalized
root causes of the
symptoms and effects

Diagnose the
dysfunctions

Address the
personalized root
causes with precision

Accessible version in your toolkit.



Differential Diagnosis of...

```
graph TD; A[Differential Diagnosis of...] --> B[Disease]; A --> C[Dysfunction];
```

Disease

Dysfunction

Clinical Focus

- Sections of the IFM Matrix.
- Types of patient data populated in each section.

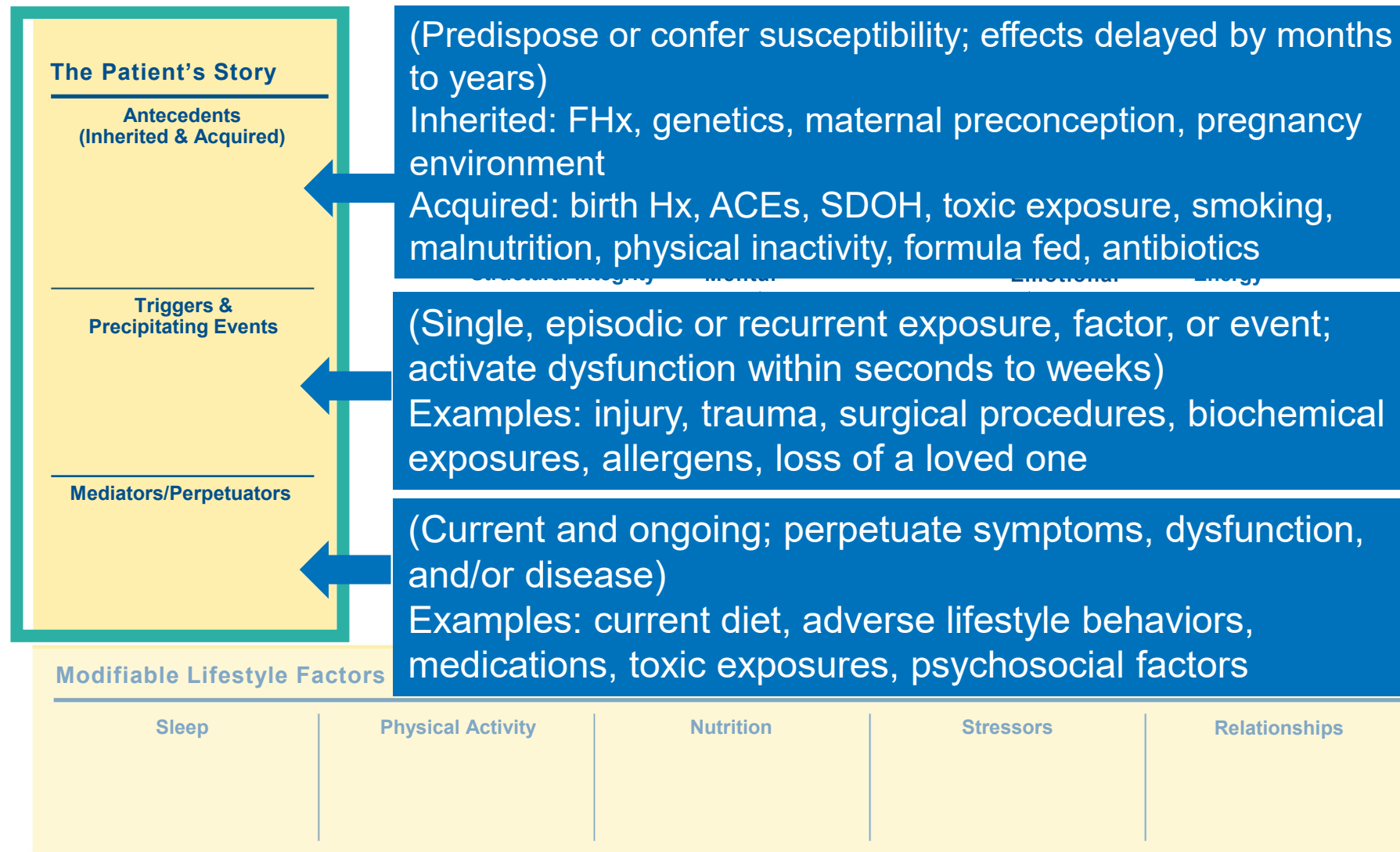


The Functional Medicine Matrix

Antecedents, Triggers, and Mediators on the **left** panel of the IFM Functional Medicine **Matrix**



Functional Medicine Matrix

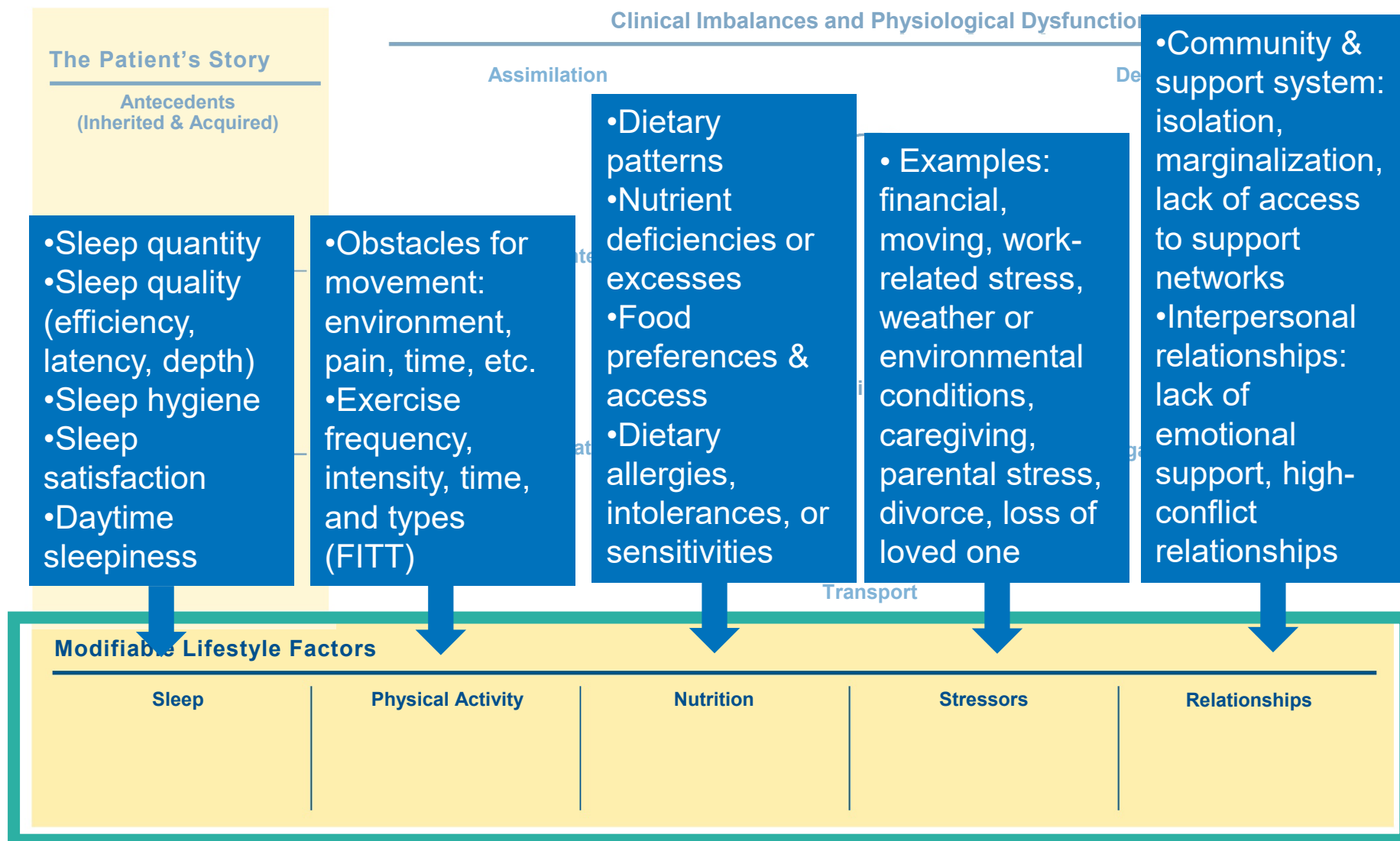


Long Description: The Functional Medicine Matrix

- The Functional Medicine Matrix
- The Patient's Story includes the following:
- Antecedents – inherited and acquired:
- Predispose or confer susceptibility; effects delayed by months to years
- Inherited: family history, genetics, maternal preconception, pregnancy, environment
- Acquired: birth history, A C Es, S D O H, toxic exposure, smoking, malnutrition, physical inactivity, formula fed
- Triggers and Precipitating Events:
- Single, episodic or recurrent exposure, factor, or event; activates dysfunction within seconds to weeks
- Examples: injury, trauma, procedure, biochemical exposures
- Mediators or Perpetuators: current and ongoing; perpetuates symptoms, dysfunction, and/or disease
- Example: current dietary pattern, lifestyle, medication, toxic exposure, psychosocial factors

Modifiable Lifestyle Factors as the **Base** of the IFM Functional Medicine **Matrix**

Functional Medicine Matrix



Long Description: The Functional Medicine Matrix

- Modifiable Lifestyle Factors
- Sleep:
 - Sleep quantity, sleep quality (efficiency, latency, depth), sleep hygiene, sleep satisfaction, daytime sleepiness
- Physical Activity:
 - Obstacles for movement: environment, pain, time, etcetera
 - Exercise frequency, intensity, time, and types (F I T T)
- Nutrition:
 - Dietary patterns, nutrient deficiencies or excess eating habits, food preferences and access, dietary allergies, intolerances, or sensitivities
- Stressors:
 - Examples: financial, moving, work related stress, weather or environmental conditions, care giving, parental stress, divorce, loss of loved one
- Relationships:
 - Community and support system: isolation, marginalization, lack of access to support networks
 - Interpersonal relationships: lack of emotional support, high-conflict relationships

Physiological Dysfunctions and Clinical Imbalances on the **Nodes** of the IFM Functional Medicine **Matrix**



Functional Medicine Matrix

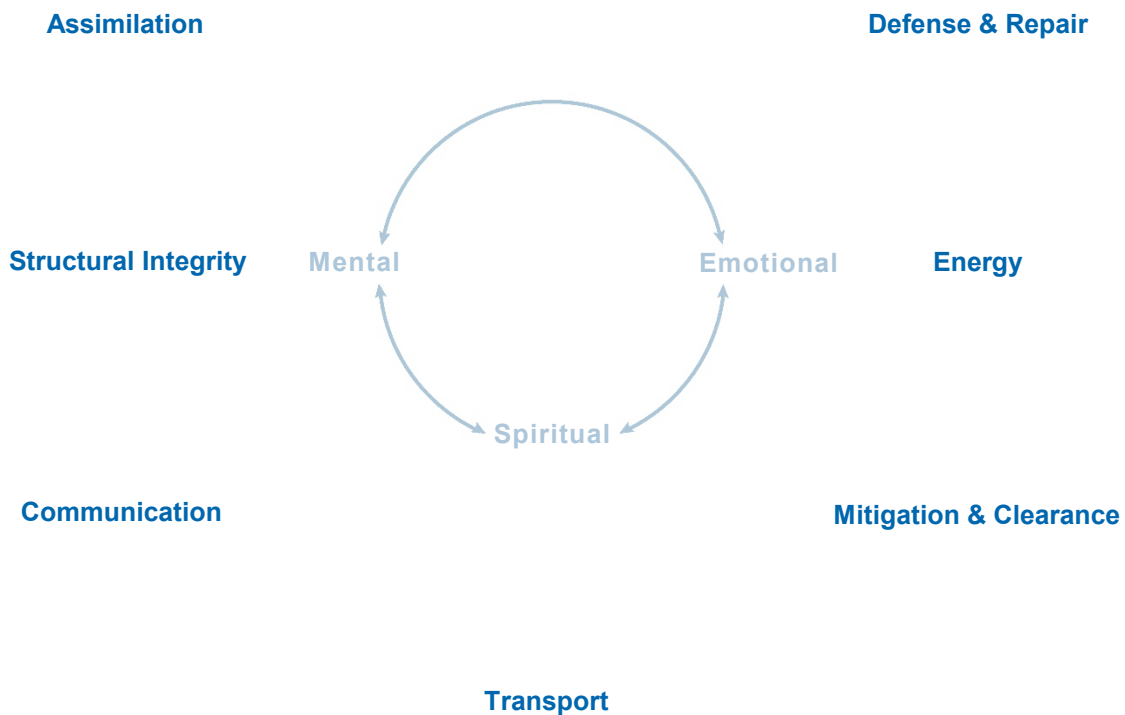
The Patient's Story

Antecedents
(Inherited & Acquired)

Triggers &
Precipitating Events

Mediators/Perpetuators

Clinical Imbalances and Physiological Dysfunctions



Modifiable Lifestyle Factors

Sleep

Physical Activity

Nutrition

Stressors

Relationships

Name: _____ Date: _____ CC: _____

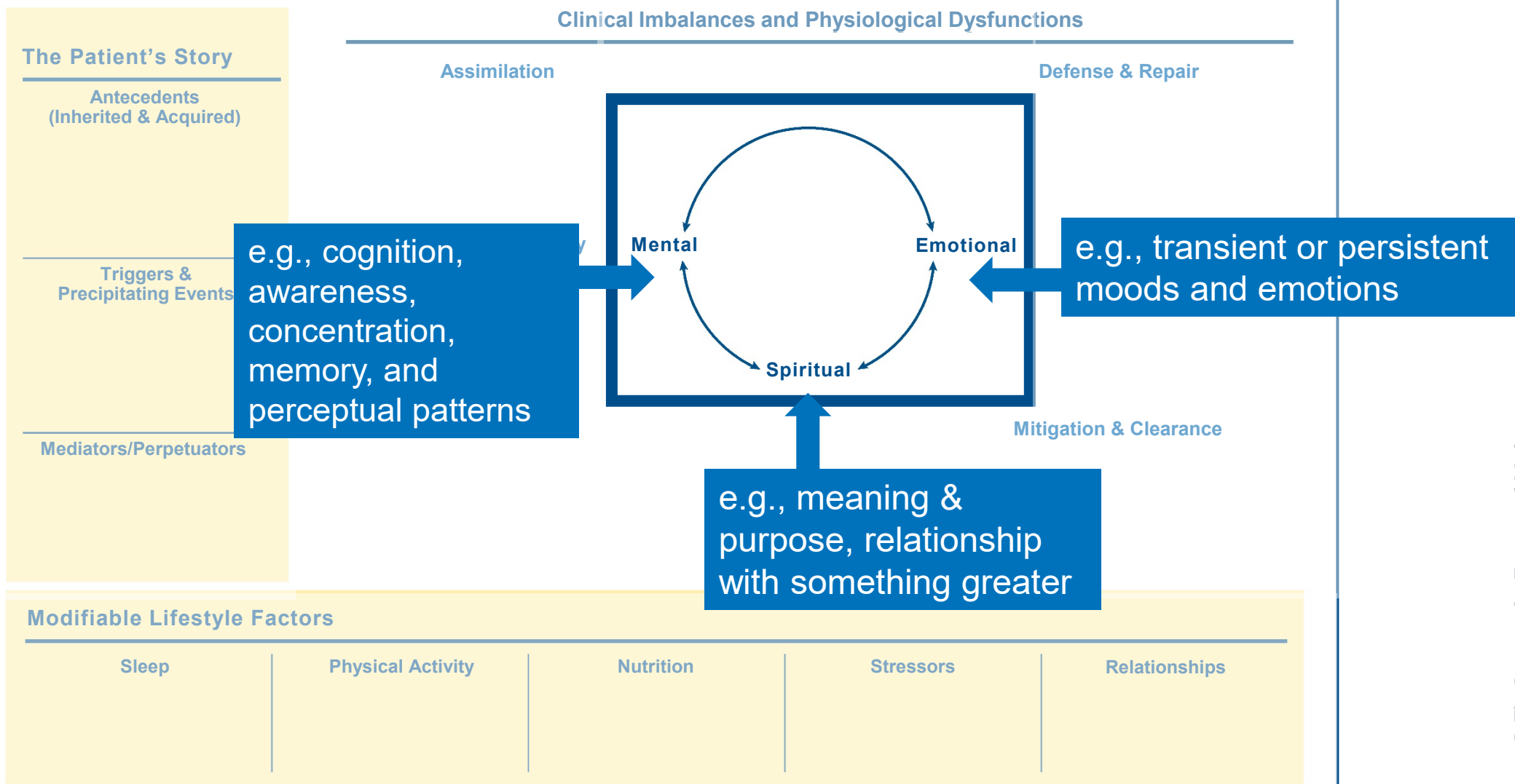
Long Description: The Functional Medicine Matrix

- Along the left-hand side of the page is the patient's story, which includes the following:
 - Antecedents – inherited and acquired
 - Triggers and precipitating events
 - Mediators and perpetuators
- Along the bottom of the page are modifiable lifestyle factors, which include the following:
 - Sleep
 - Physical activity
 - Nutrition
 - Stressors
 - Relationships
- In the center of the page are clinical imbalances and physiological dysfunctions
 - These nodes include assimilation, defense and repair, structural integrity, energy, communication, mitigation and clearance, and transport
 - Within the center is a circle that includes mental, emotional and spiritual

Mental, Emotional, & Spiritual Factors in the **Center** of the IFM Functional Medicine **Matrix**



Functional Medicine Matrix



Long Description: The Functional Medicine Matrix

- The Functional Medicine Matrix highlighting the mental, emotional, and spiritual factors in the center of the matrix.
- Mental factors include cognition, awareness, concentration, memory, and perceptual patterns
- Emotional factors include transient or persistent moods and emotions
- Spiritual factors include meaning and purpose, relationship with something greater

Clinical Focus

- Description of the physiological functions represented in each node of the IFM Matrix.
- Interrelationships among the functions in each of the nodes.



Functional Medicine Matrix

The Patient's Story

Antecedents
(Inherited & Acquired)

Triggers &
Precipitating Events

Mediators/Perpetuators

Modifiable Lifestyle Factors

Sleep

Physical Activity

Nutrition

Stressors

Relationships

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity

Mental

Emotional

Energy

Spiritual

Communication

Mitigation & Clearance

Transport

Assimilation

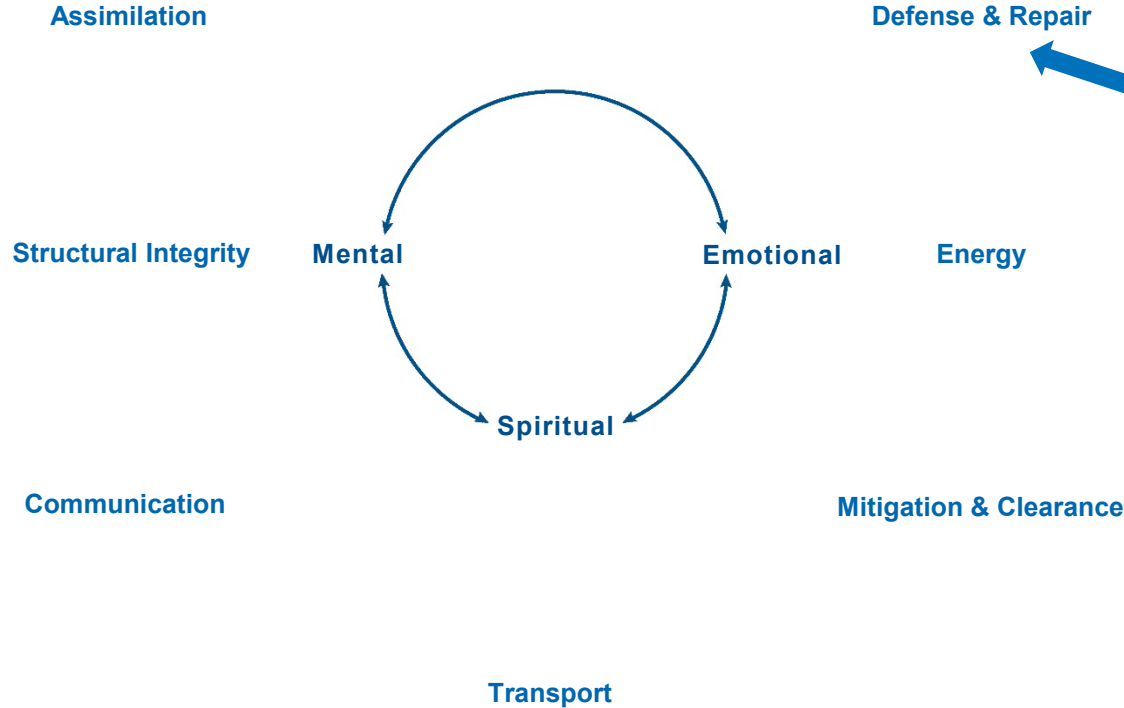
(e.g., digestion,
absorption,
microbiota/GI,
respiration)

Long Description: The Functional Medicine Matrix

- The Functional Medicine Matrix emphasizing the assimilation node
- Examples include digestion, absorption, microbiota/GI, respiration

Functional Medicine Matrix

Clinical Imbalances and Physiological Dysfunctions



Defense and Repair
(e.g., immune, inflammation, infection/microbiota)

The Patient's Story

Antecedents
(Inherited & Acquired)

Triggers & Precipitating Events

Mediators/Perpetuators

Modifiable Lifestyle Factors

Sleep

Physical Activity

Nutrition

Stressors

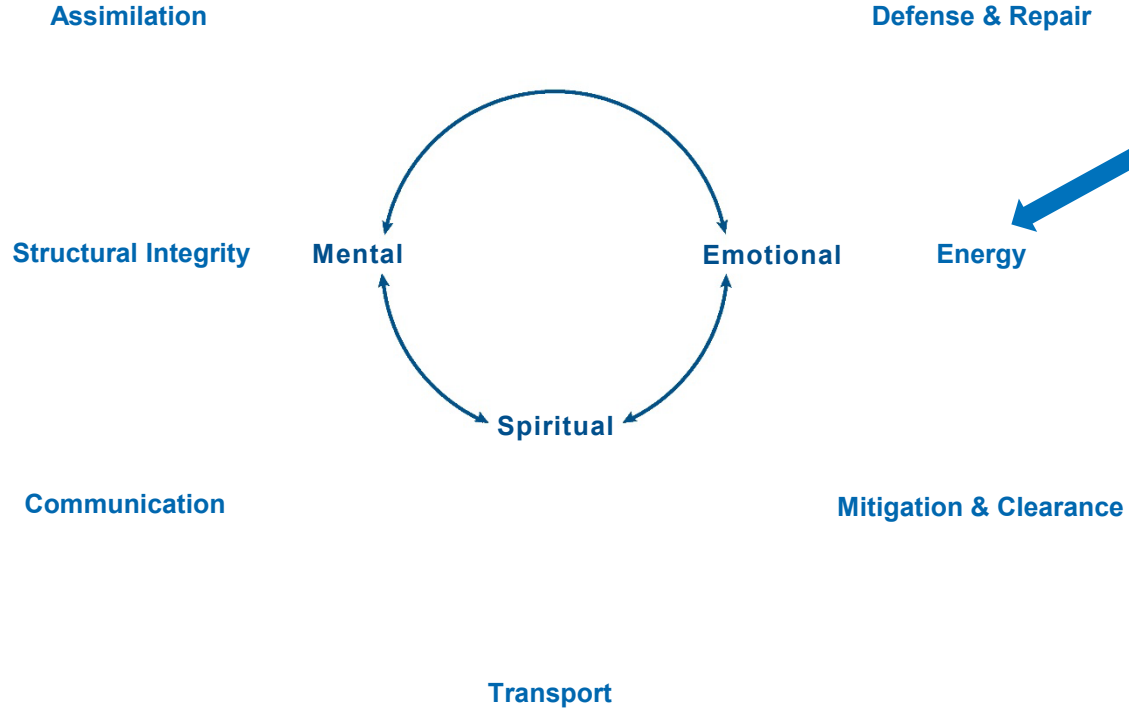
Relationships

Long Description: The Functional Medicine Matrix

- The Functional Medicine Matrix emphasizing the defense and repair node
- Examples include immune, inflammation, infection/microbiota

Functional Medicine Matrix

Clinical Imbalances and Physiological Dysfunctions



The Patient's Story

Antecedents
(Inherited & Acquired)

Triggers & Precipitating Events

Mediators/Perpetuators

Modifiable Lifestyle Factors

Sleep

Physical Activity

Nutrition

Stressors

Relationships

Long Description: The Functional Medicine Matrix

- The Functional Medicine Matrix emphasizing the energy node
- Examples include energy regulation, mitochondrial function

Functional Medicine Matrix

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity

Mental

Emotional

Energy

Spiritual

Communication

Mitigation & Clearance

Transport

**Mitigation and
Clearance**
(e.g., toxic load,
detoxification)

The Patient's Story

Antecedents
(Inherited & Acquired)

Triggers &
Precipitating Events

Mediators/Perpetuators

Modifiable Lifestyle Factors

Sleep

Physical Activity

Nutrition

Stressors

Relationships

Long Description: The Functional Medicine Matrix

- The Functional Medicine Matrix emphasizing the mitigation and clearance node
- Examples include toxic load, detoxification

Functional Medicine Matrix

Clinical Imbalances and Physiological Dysfunctions

The Patient's Story

Antecedents
(Inherited & Acquired)

Triggers &
Precipitating Events

Mediators/Perpetuators

Assimilation

Defense & Repair

Structural Integrity

Mental

Emotional

Energy

Spiritual

Communication

Mitigation & Clearance

Transport

Transport
(e.g., cardiovascular,
lymphatic system)

Modifiable Lifestyle Factors

Sleep

Physical Activity

Nutrition

Stressors

Relationships

Long Description: The Functional Medicine Matrix

- The Functional Medicine Matrix emphasizing the transport node
- Examples include cardiovascular, lymphatic system

Functional Medicine Matrix

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity

Mental

Emotional

Energy

Spiritual

Communication

Mitigation & Clearance

Transport

Communication

(e.g., endocrine,
neurotransmitters,
immune messengers)

The Patient's Story

Antecedents
(Inherited & Acquired)

Triggers &
Precipitating Events

Mediators/Perpetuators

Modifiable Lifestyle Factors

Sleep

Physical Activity

Nutrition

Stressors

Relationships

Long Description: The Functional Medicine Matrix

- The Functional Medicine Matrix emphasizing the communication node
- Examples include endocrine, neurotransmitters, immune messengers

Functional Medicine Matrix

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity

Mental

Emotional

Energy

Spiritual

Communication

Mitigation & Clearance

Transport

Structural Integrity
(e.g., from subcellular membranes to musculoskeletal structure)

The Patient's Story

Antecedents
(Inherited & Acquired)

Triggers &
Precipitating Events

Mediators/Perpetuators

Modifiable Lifestyle Factors

Sleep

Physical Activity

Nutrition

Stressors

Relationships

Long Description: The Functional Medicine Matrix

- The Functional Medicine Matrix emphasizing the structural integrity node
- Examples include from subcellular membranes to musculoskeletal structure



Functional Medicine Matrix

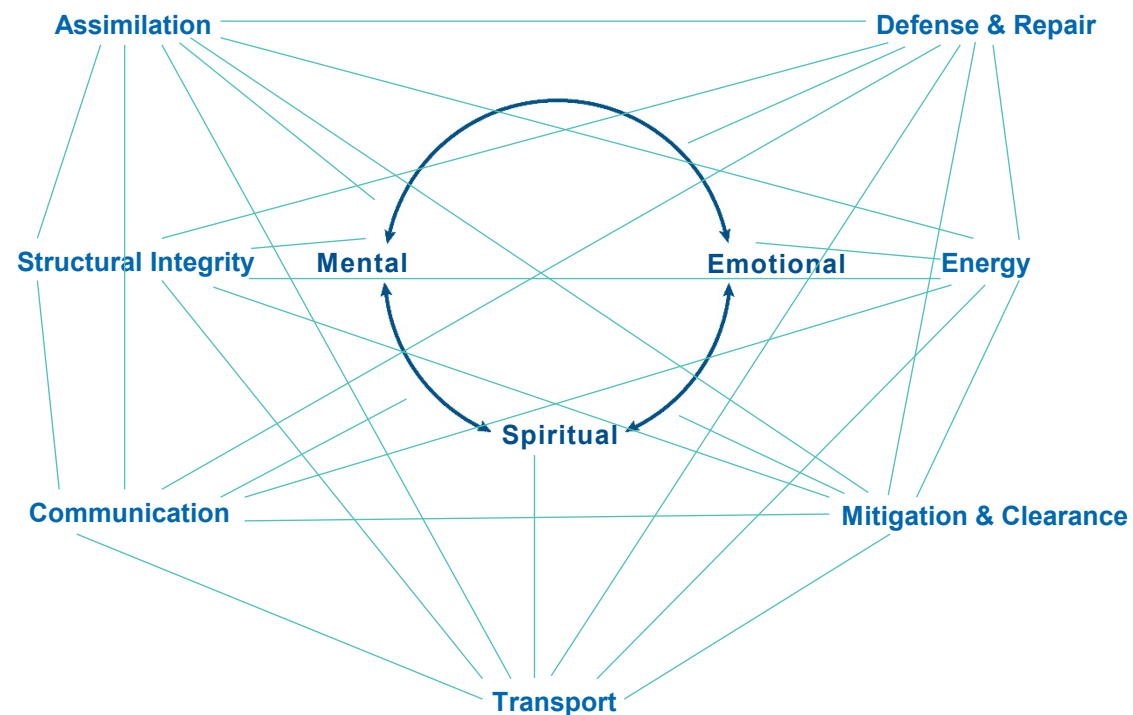
The Patient's Story

Antecedents
(Inherited & Acquired)

Triggers &
Precipitating Events

Mediators/Perpetuators

Clinical Imbalances and Physiological Dysfunctions



Modifiable Lifestyle Factors

Sleep

Physical Activity

Nutrition

Stressors

Relationships

Name: _____ Date: _____ CC: _____

Long Description: The Functional Medicine Matrix

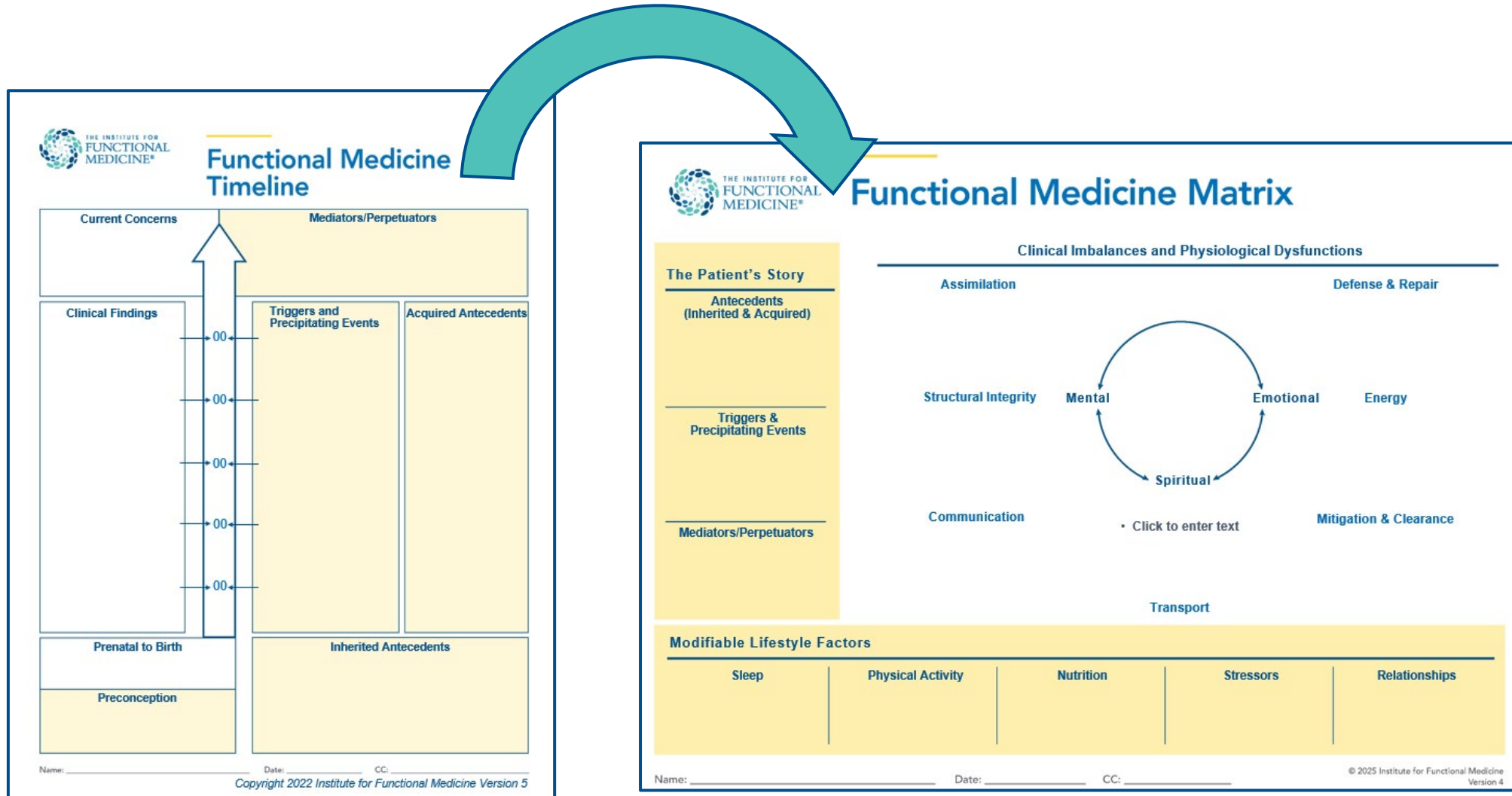
- The Functional Medicine Matrix with arrows connecting each node

Clinical Focus

- How to populate the matrix.
- Transposing data from the timeline to the matrix.
- Patient data on the matrix that is not on the timeline.
- Populating patient data in more than one section of the matrix.



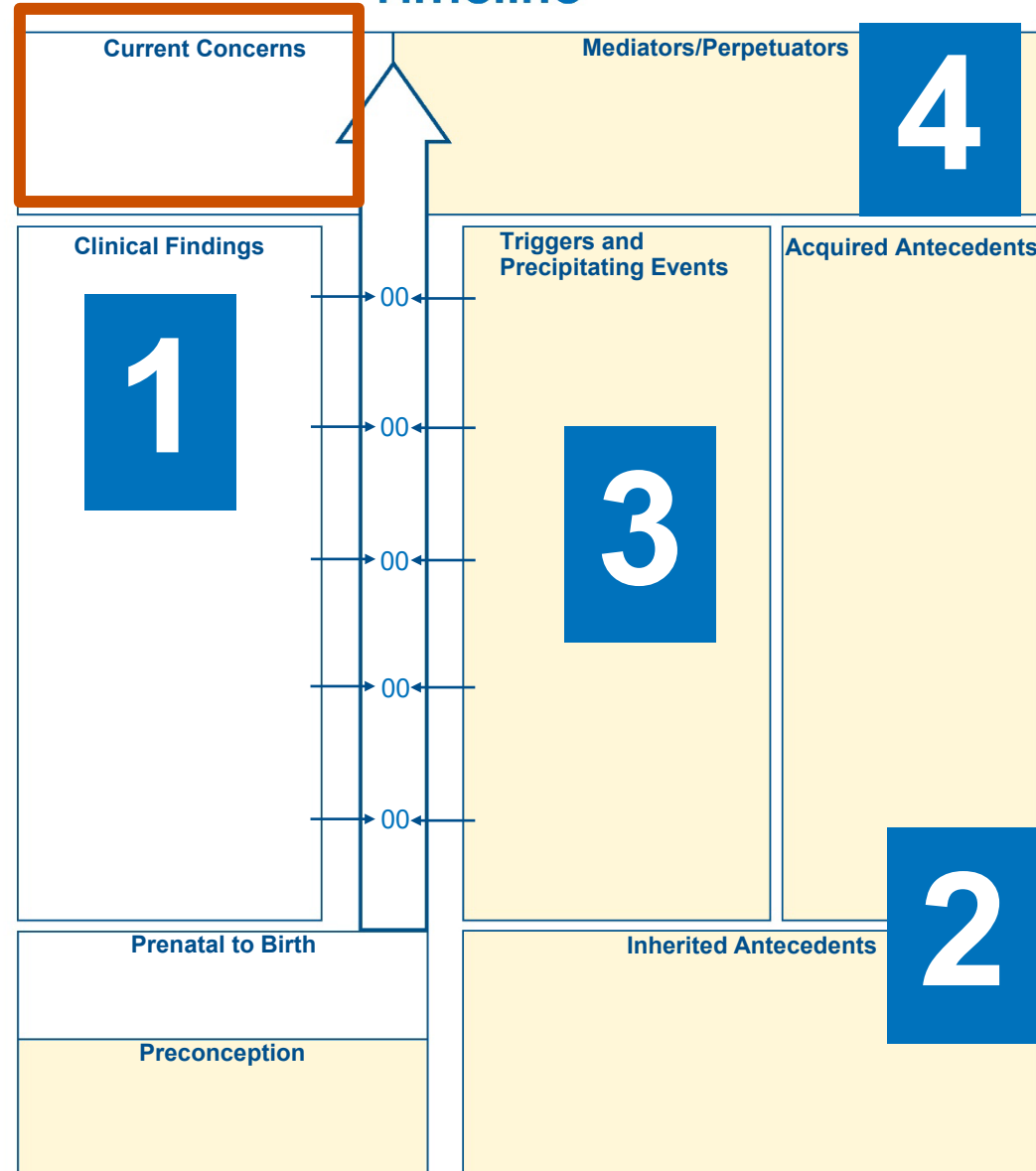
Transferring Clinical Findings, ATMs, and MLFs from the Timeline to the Matrix



Long Description: Transferring Clinical Findings, ATMs, and MLFs from the Timeline to the Matrix

- An arrow pointing from a blank Functional Medicine Timeline to a blank Functional Medicine Matrix.

Functional Medicine Timeline



Name: _____

Date: _____

CC: _____

Long Description: The Functional Medicine Timeline

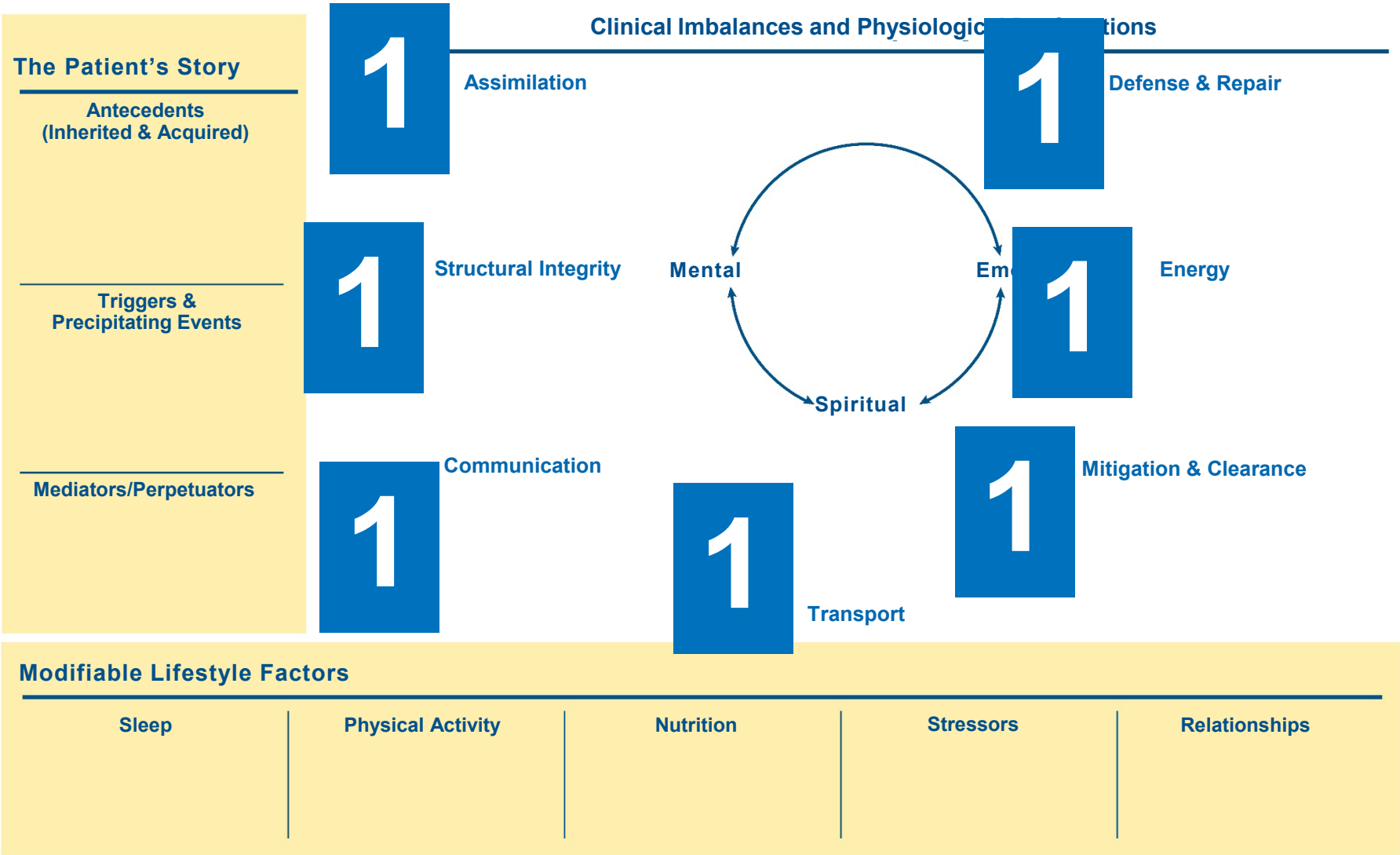
- The Functional Medicine Timeline
 1. Clinical Findings
 2. Preconception and Inherited Antecedents
 3. Triggers and Precipitating Events, Acquired Antecedents
 4. Mediators and Perpetuators

SEQUENCE FOR
TRANSPOSING
TIMELINE
FACTORS TO
THE MATRIX
**CLINICAL
FINDINGS**

Keep in mind that clinical findings represent the observable “effects” of the root causes of the patient’s conditions/concerns.



Functional Medicine Matrix

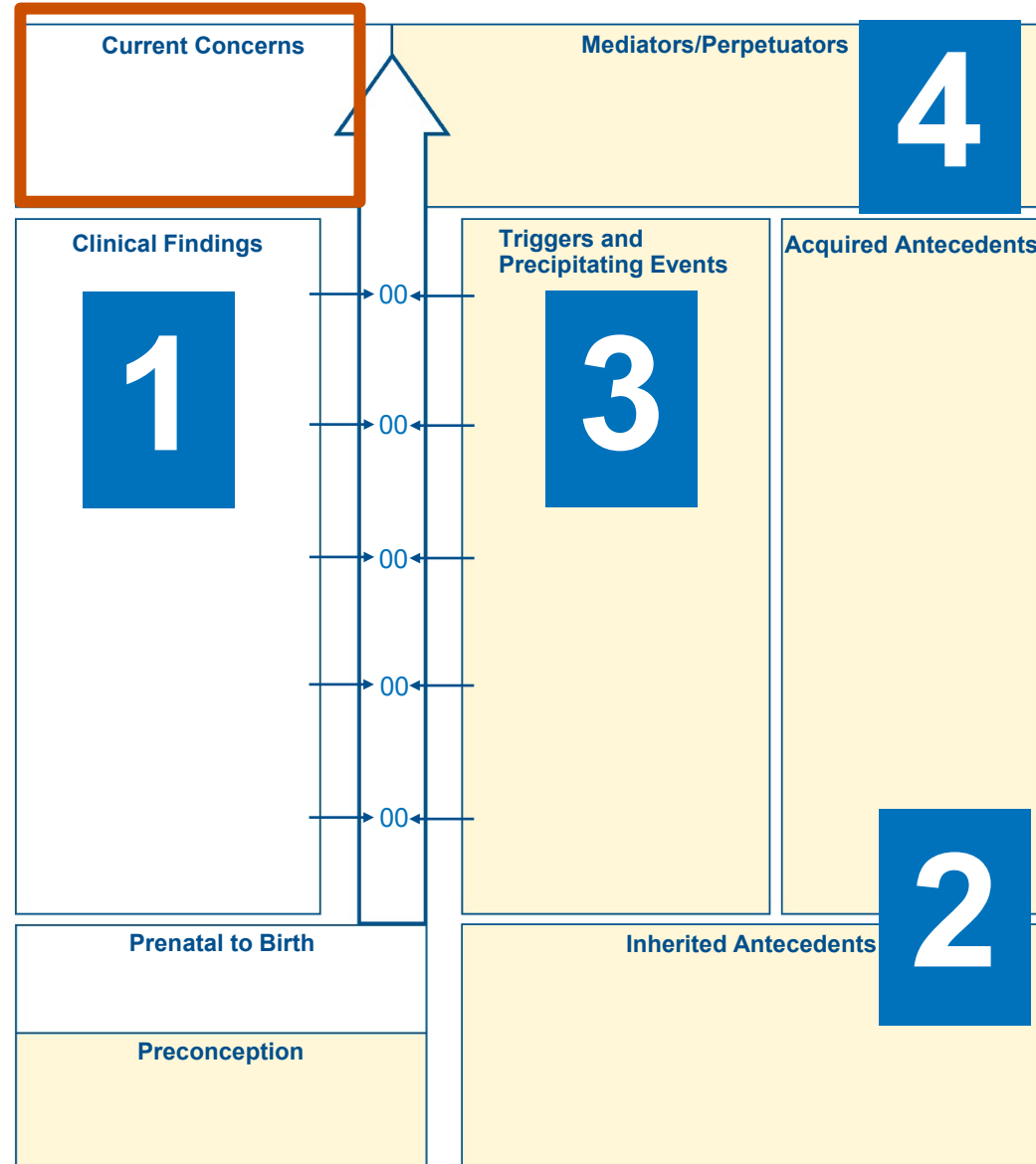


Name: _____ Date: _____ CC: _____

Long Description: The Functional Medicine Matrix

- The Functional Medicine Matrix with an emphasis on the clinical imbalances and physiological dysfunctions
- Each node is assigned a number one.
- Clinical findings from the Functional Medicine Timeline can be placed in these nodes.

Functional Medicine Timeline



Name: _____ Date: _____ CC: _____

Long Description: The Functional Medicine Timeline

- The Functional Medicine Timeline
 1. Clinical Findings
 2. Preconception and Inherited Antecedents
 3. Triggers and Precipitating Events, Acquired Antecedents
 4. Mediators and Perpetuators

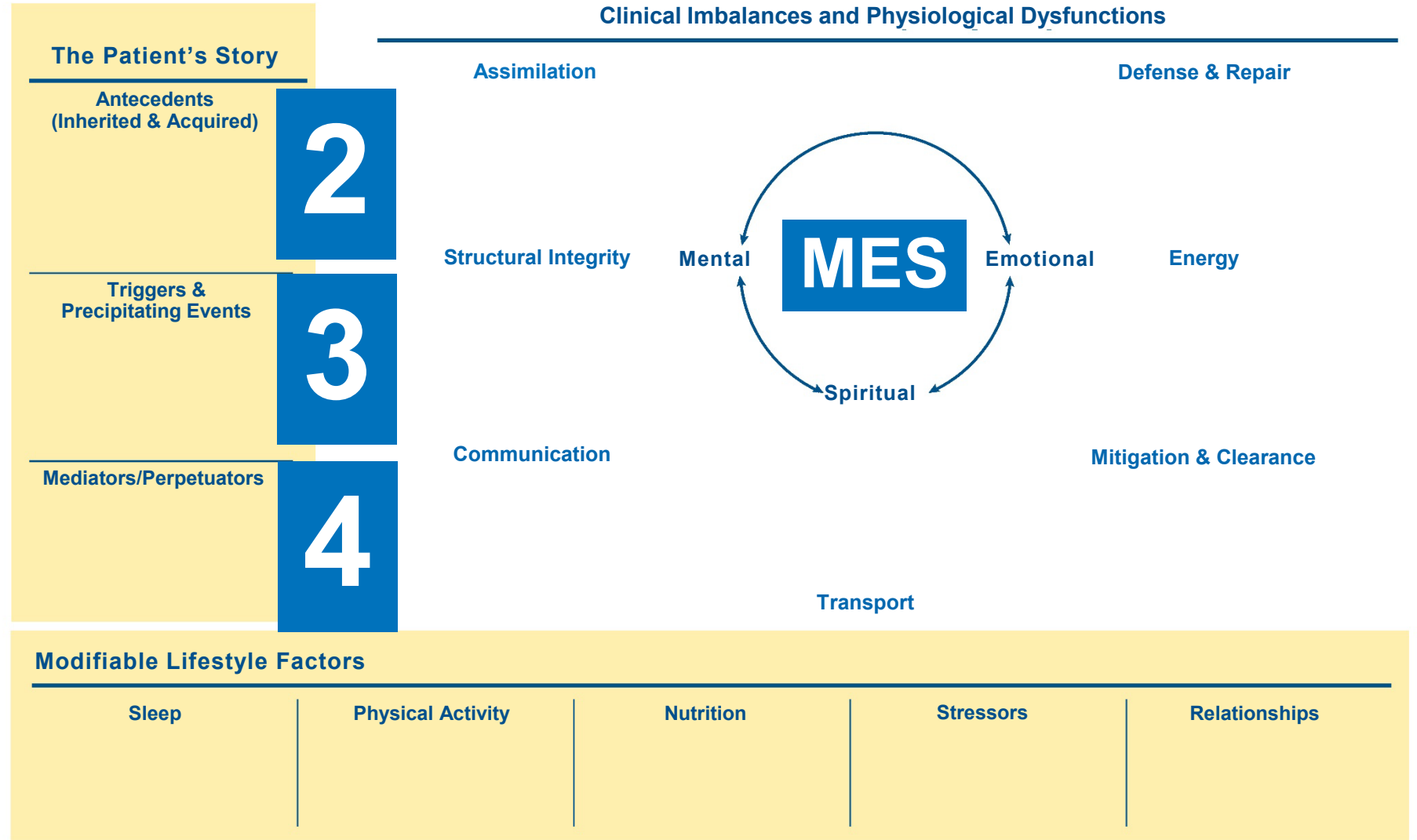
SEQUENCE FOR TRANSPOSING TIMELINE FACTORS TO THE MATRIX ANTECEDENTS TRIGGERS MEDIATORS M/E/S FACTORS

Keep in mind that all mental, emotional, and spiritual factors may not have been represented on the timeline, but they are essential to populate in the center of the matrix.



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Functional Medicine Matrix



Name: _____ Date: _____ CC: _____

Long Description: The Functional Medicine Matrix

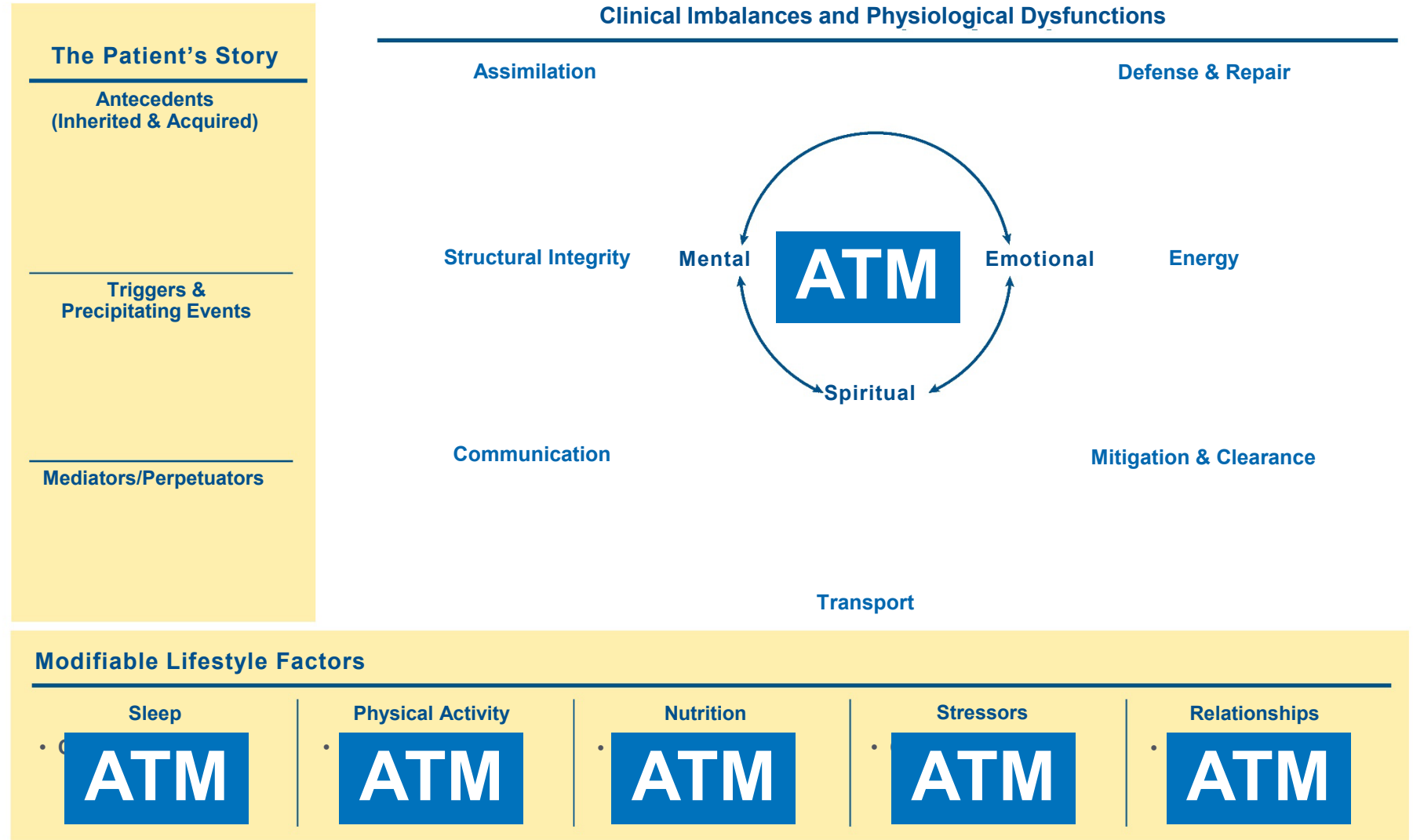
- The Functional Medicine Matrix with an emphasis on the patient's story.
- 2: Antecedents – inherited and acquired
- 3: Triggers and Precipitating Events
- 4: Mediators and Perpetuators
- Clinical findings from the Functional Medicine Timeline can be placed in this area of the matrix based on the corresponding numbers.

Duplication of factors populated on the matrix

Keep in mind that all mental, emotional, spiritual, dietary, and modifiable lifestyle factors may also be antecedents, triggers, and/or mediators of physiological dysfunction.



Functional Medicine Matrix



Name: _____ Date: _____ CC: _____

Long Description: The Functional Medicine Matrix

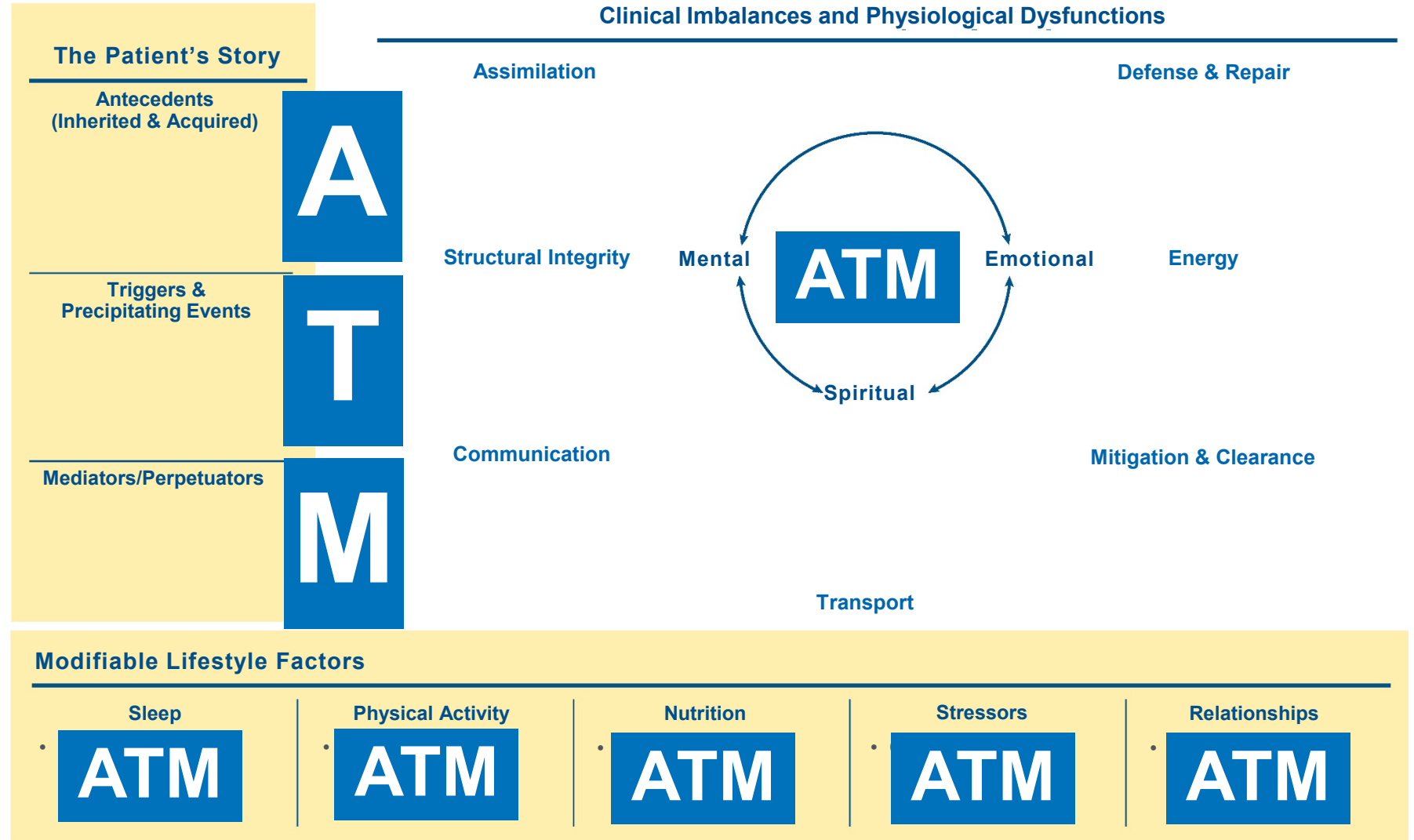
- The Functional Medicine Matrix with an emphasis on M E S and modifiable lifestyle factors. M E S and all modifiable lifestyle factors are labeled as A T Ms.

Duplication of factors populated on the matrix

Therefore, any of these individual factors may be populated in more than one section of the matrix.



Functional Medicine Matrix



Name: _____ Date: _____ CC: _____

Long Description: The Functional Medicine Matrix

- The Functional Medicine Matrix with an emphasis on M E S, modifiable lifestyle factors, and the patient's story. M E S, all modifiable lifestyle factors, and the patient's story are labeled as A T Ms.

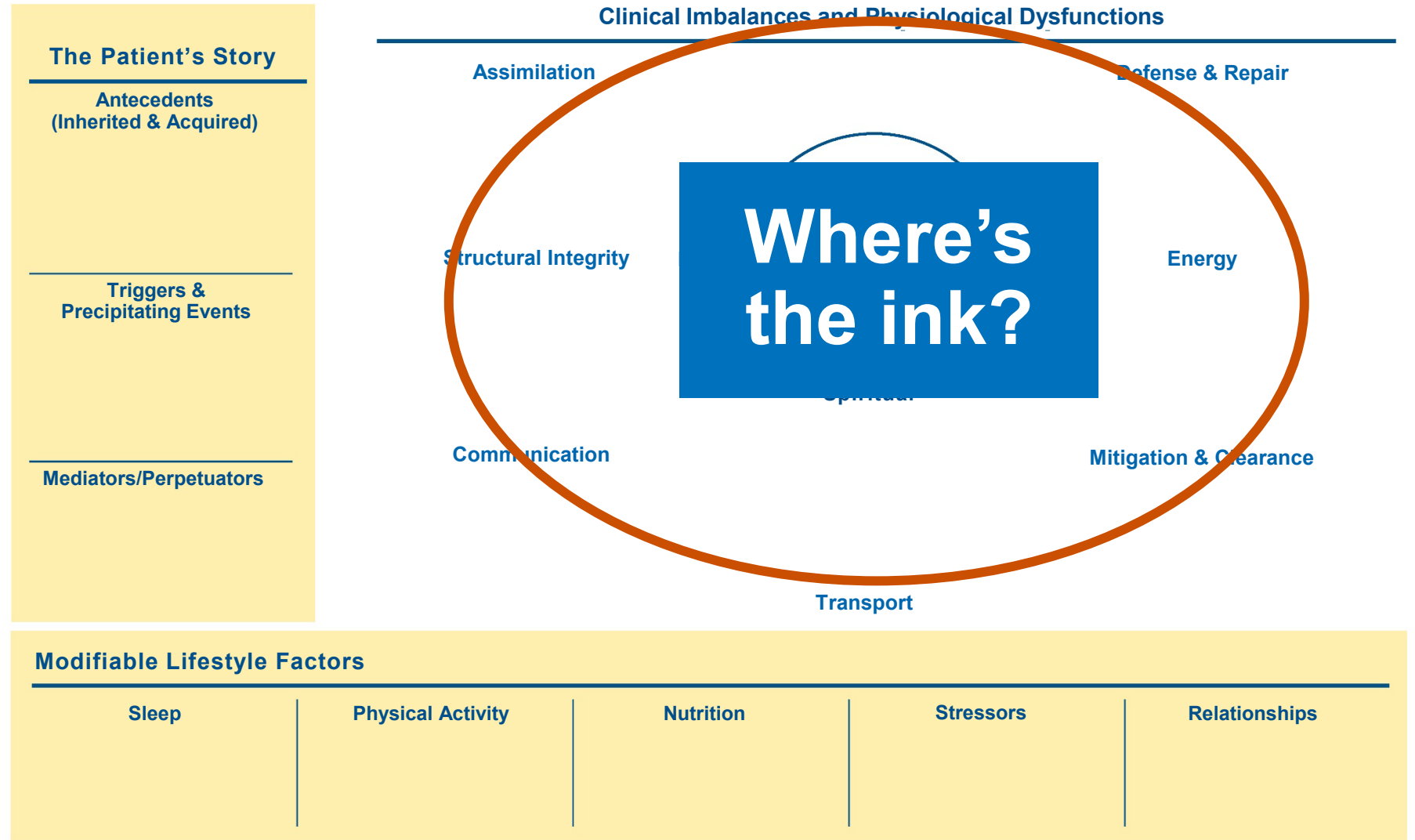
Where's the ink?

The distribution and pattern of the clinical findings populated on the matrix nodes assist you in generating an **explanation** of the individualized root causes of the patient's conditions and concerns.

You will “Re-Tell” this explanation (story) to the patient in language they can understand in step 3 of GOTOIT.



Functional Medicine Matrix



Name: _____ Date: _____ CC: _____

Long Description: The Functional Medicine Matrix

- The Functional Medicine Matrix with a red circle around the Clinical Imbalances and Physiological Dysfunctions. The question – Where's the link? – is in the center of the circle overlying M E S.

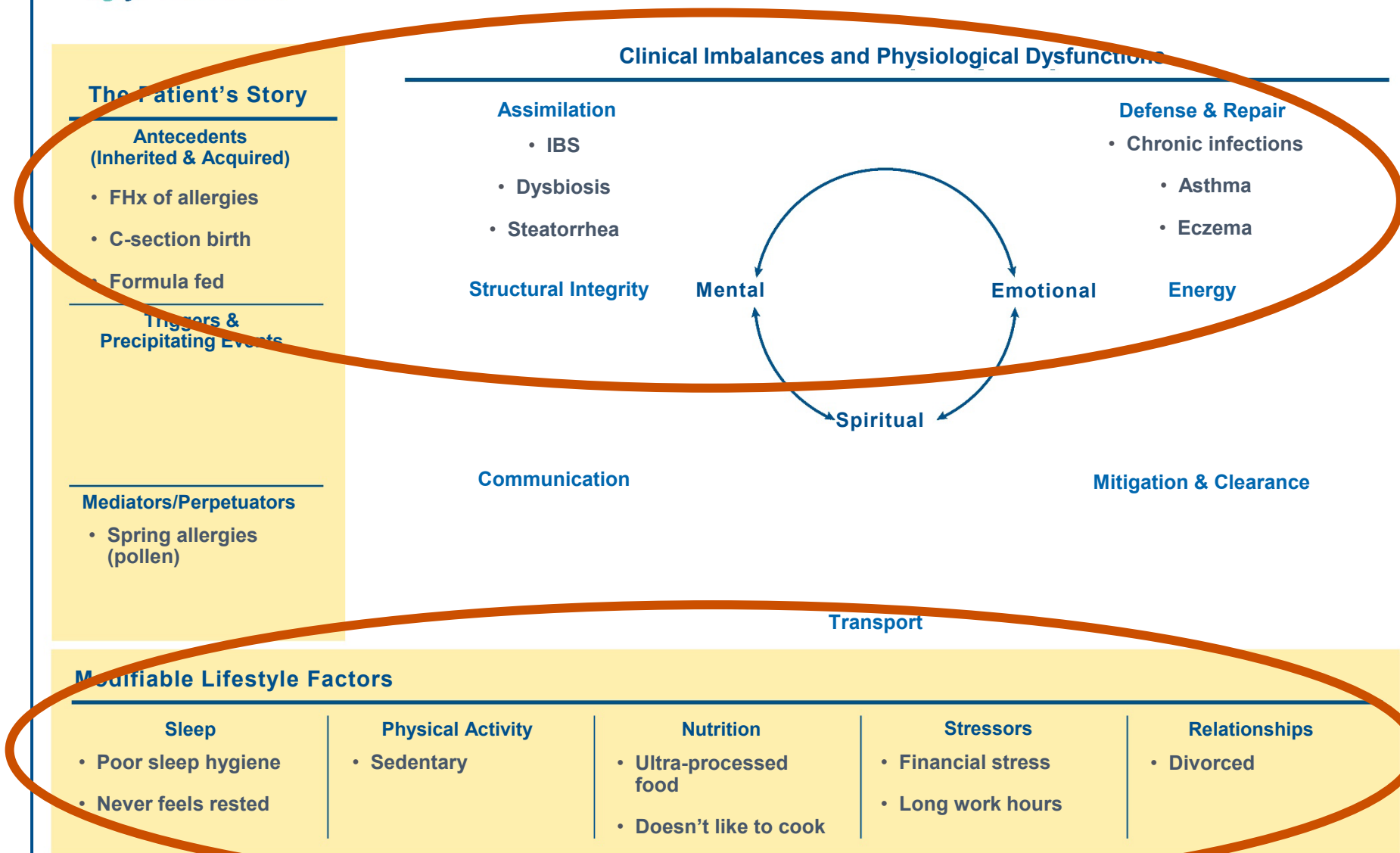
How should you clinically apply the nodes of the matrix?

Additionally, the distribution and pattern of clinical findings as they are populated on the nodes of the matrix provide the basis, and the “personalized lines of evidence,” for prioritizing evaluative and therapeutic interventions in the fourth GOTOIT step.



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Functional Medicine Matrix



Name: _____ Date: _____

Long Description: The Functional Medicine Matrix

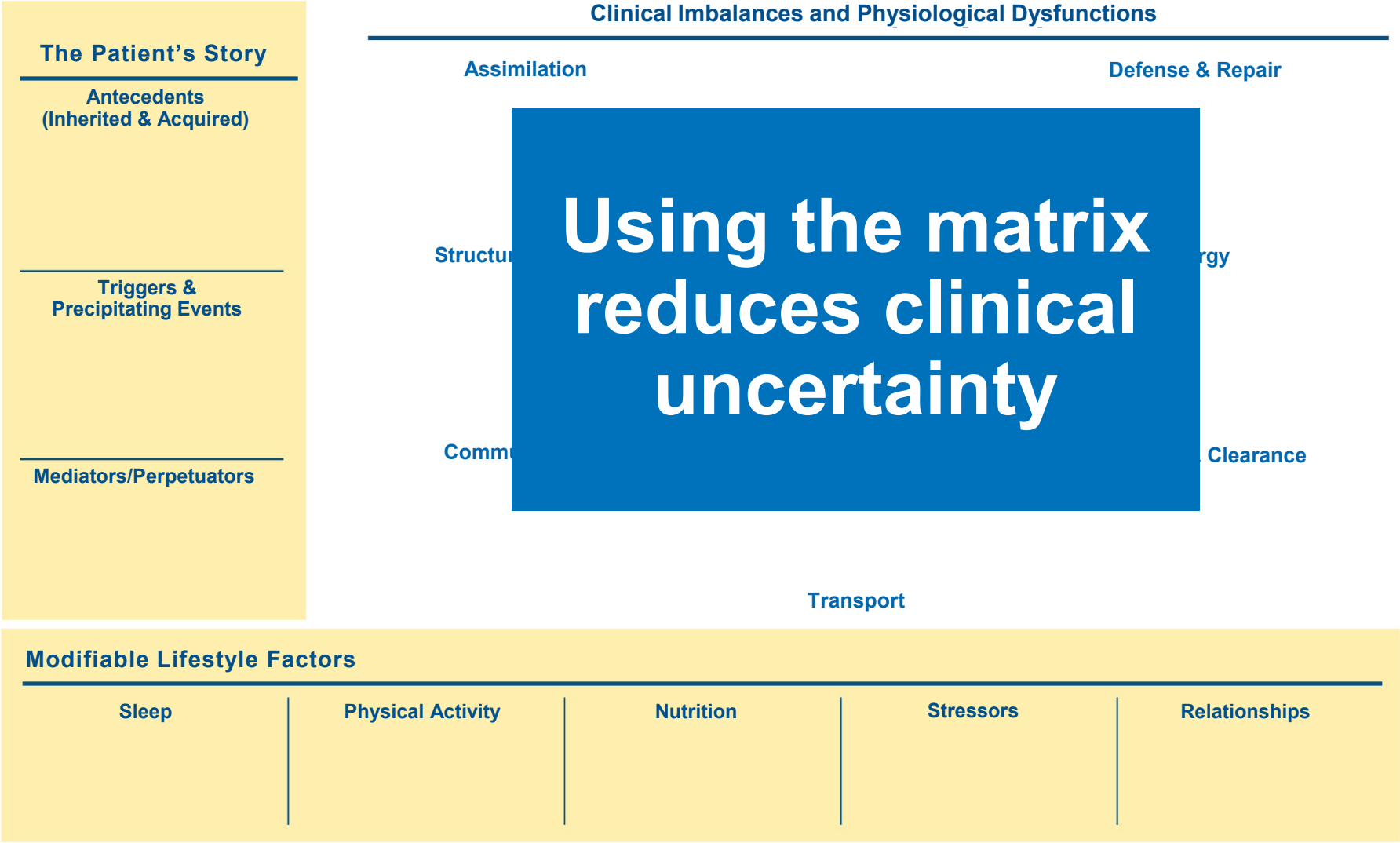
- An example of a Functional Medicine Matrix completed with patient information.
- The Patient's Story
- Antecedents – Inherited and Acquired
- Family history of allergies
- C-section birth
- Formula fed
- Mediators and perpetuators
- Spring allergies (pollen)
- Modifiable lifestyle factors
- Sleep
- Poor sleep hygiene
- Never feels rested
- Physical Activity
- Sedentary
- Nutrition
- Ultra-processed food
- Doesn't like to cook
- Stressors
- Financial stress
- Long work hours
- Relationships
- Divorced
- One red circle around the patient's story and clinical imbalances and physiological dysfunctions and a second circle around the modifiable lifestyle factors highlights how to apply the clinical findings in the Matrix nodes to the G O T O I T model.

Functional medicine reduces uncertainty in clinical practice.

Use of the matrix reduces uncertainty in clinical practice. With skillful interpretation, it informs personalized and precision-based recommendations for each patient based on their unique set of root cause contributors.



Functional Medicine Matrix



Long Description: The Functional Medicine Matrix

- A blank Functional Medicine Matrix with the statement – Using the Matrix reduces clinical uncertainty – over the clinical imbalances and physiological dysfunctions.

Functional Medicine Matrix (Teaching)

The Patient's Story

Antecedents (Inherited & Acquired)

(Predispose or confer susceptibility; effects delayed by months to years)

- Inherited: FHx, genetics, maternal preconception, pregnancy environment
- Acquired: birth Hx, ACEs, SDOH, toxic exposure, smoking, malnutrition, physical inactivity, formula fed

Triggers & Precipitating Events

(Single, episodic or recurrent exposure, factor, or event; activates dysfunction within seconds to weeks)

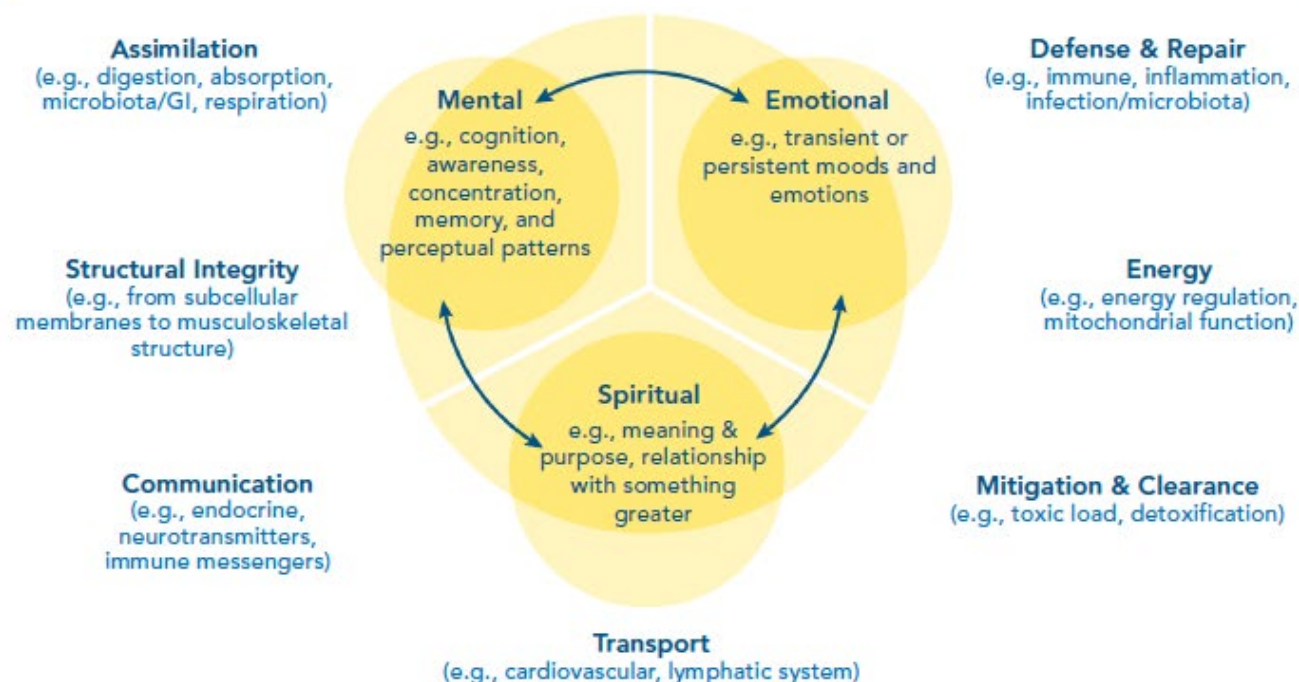
- Examples: injury, trauma, procedure, biochemical exposures

Mediators/Perpetuators

(Current and ongoing; perpetuates symptoms, dysfunction, and/or disease)

- Examples: current dietary pattern, lifestyle, medication, toxic exposures, psychosocial factors

Clinical Imbalances and Physiological Dysfunctions



Modifiable Lifestyle Factors (Focus on pertinent negatives for potential modification. Consider coping mechanisms related to MES)

Sleep

- Sleep quantity
- Sleep quality (efficiency, latency, depth)
- Sleep hygiene
- Sleep satisfaction
- Daytime sleepiness

Physical Activity

- Obstacles for movement: environment, pain, time, etc.
- Exercise frequency, intensity, time, and types (FITT)

Nutrition

- Dietary patterns
- Nutrient deficiencies or excess eating habits
- Food preferences & access
- Dietary allergies, intolerances, or sensitivities

Stressors

- Examples: financial, moving, work related stress, weather or environmental conditions, care giving, parental stress, divorce, loss of loved one

Relationships

- Community & support system: isolation, marginalization, lack of access to support networks
- Interpersonal relationships: lack of emotional support, high-conflict relationships

Long Description: The Functional Medicine Matrix (Teaching)

- The Functional Medicine Matrix Teaching Example
- The Patient's Story
- Antecedents – inherited and acquired
- Predispose or confer susceptibility; effects delayed by months to years
- Inherited: family history, genetics, maternal preconception, pregnancy, environment
- Acquired: birth history, A C Es, S D O H, toxic exposure, smoking, malnutrition, physical inactivity, formula fed
- Triggers and Precipitating Events
- Single, episodic, or recurrent exposure, factor, or event; activates dysfunction within seconds to weeks
- Examples include injury, trauma, procedure, biochemical exposures
- Mediators or perpetuators
- Current and ongoing: perpetuates symptoms, dysfunction and or disease
- Examples include current dietary pattern, lifestyle, medication, toxic exposures, psychosocial factors
- Modifiable Lifestyle Factors
- Focus on pertinent negatives for potential modification. Consider coping mechanisms related to M E S.
- Sleep:
- Sleep quantity, sleep quality (efficiency, latency, depth), sleep hygiene, sleep satisfaction, daytime sleepiness
- Physical Activity:
- Obstacles for movement: environment, pain, time, etcetera
- Exercise frequency, intensity, time, and types (F I T T)
- Nutrition:
- Dietary patterns, nutrient deficiencies or excess eating habits, food preferences and access, dietary allergies, intolerances, or sensitivities
- Stressors:
- Examples: financial, moving, work related stress,

Clinical Focus

- Analyzing patterns on the matrix to generate an explanation of the root causes of the patient's condition.



Generation of the Explanation:

Three Common Patient Profiles

- Patient A who feels well and wants to maintain wellness.
- Patient B with multiple vague “undiagnosable” symptoms.
- Patient C with multiple common chronic diseases (NCDs).

Generation of the Explanation:

Wellness, Prevention, Preclinical Conditions

- Patient A who feels well and wants to maintain wellness:
 - This patient's ATMs, MLFs, and MES factors will help you generate an explanation of the most likely **silent, incipient, and nascent physiological dysfunctions** and clinical imbalances that may be **identified, mitigated, and reversed in their preclinical stages.**
 - “Precision-based prognostic and preventive personalized root cause medicine.”

Patient A

Sofia, 39 years old

ATMs

Fam Hx: T2DM

Fam Hx: CAD

Chronic stress



ATMs

Hx OCP use

Early menopause

Premature puberty

Generation of the Explanation: **Vague, Syndromic, Undiagnosable Symptoms**

- **Patient B** with multiple vague “undiagnosable” symptoms:
 - The distribution of this patient’s **symptoms** and other **non-diagnostic clinical findings** on the nodes of the matrix will help you generate an explanation of the most significant root causes of their physiological dysfunctions and clinical imbalances, even when a diagnosis remains elusive.
 - **“Precision-based personalized root cause medicine.”**

Patient B

Sofia, 39 years old

Current Concerns

Fatigue

Arthralgias

Brain fog



Current Concerns

Works as a dry cleaner

Tobacco dependence

Adverse drug reactions

Generation of the Explanation: **Multiple Chronic Diseases**

- **Patient C** with multiple common chronic diseases (NCDs):
 - The distribution of this patient's **known diagnoses** on the nodes of the matrix will help you recognize **patterns** and generate an explanation of the most significant **root causes** of physiological dysfunction and clinical imbalances that underpin their diagnoses.
 - “**Precision-based personalized root cause medicine.**”

Patient C

Sofia, 39 years old

Current Concerns

Dyspepsia

Diarrhea

PPI

Diagnoses

GERD

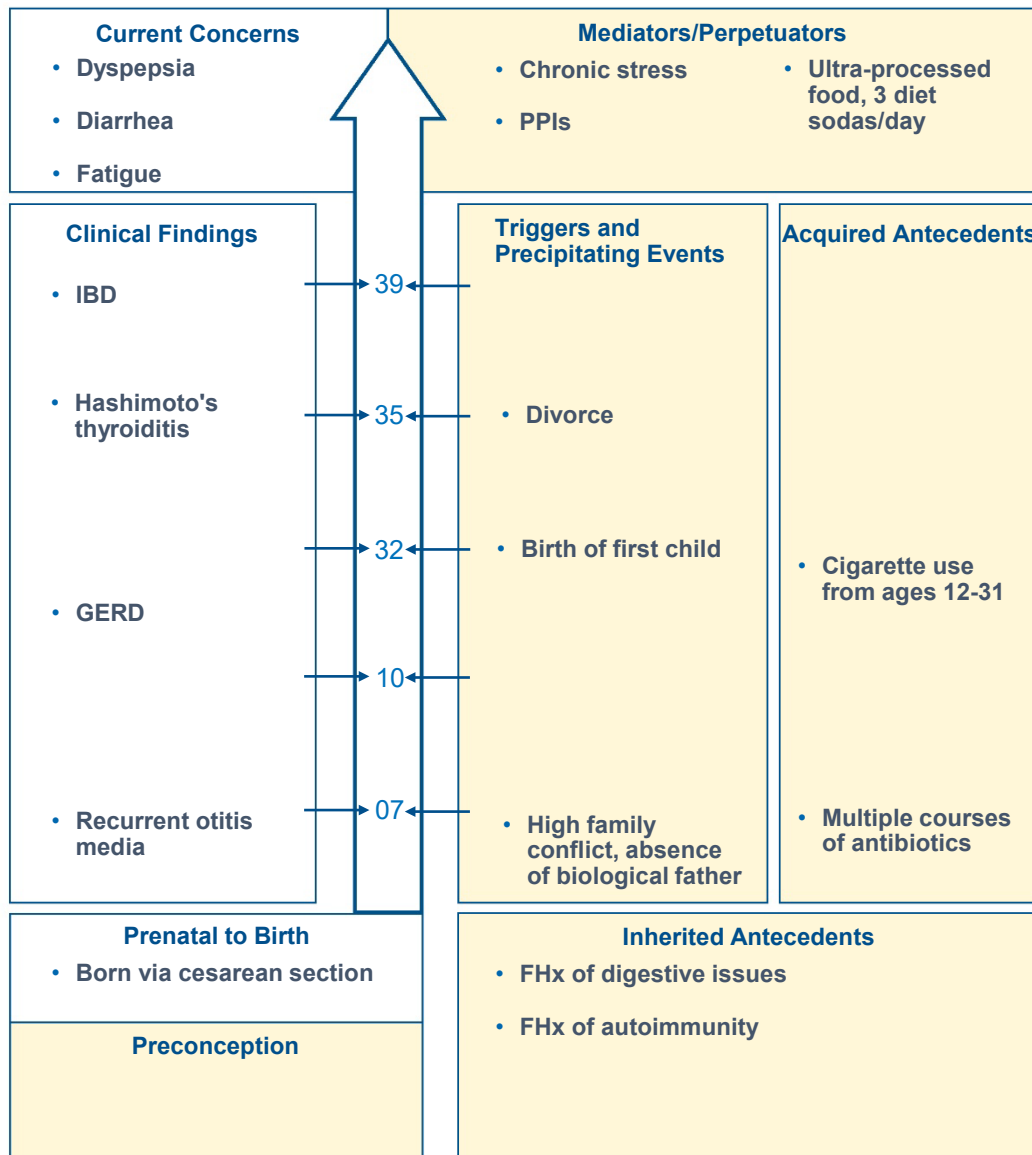
Hashimoto's

IBD



Functional Medicine Timeline

Patient C
Sofia
Age 39



Name: _____ Date: _____ CC: _____

Long Description: Functional Medicine Timeline

- The Functional Medicine Timeline for Patient C
- Current concerns
- Dyspepsia, diarrhea, fatigue
- Inherited antecedents
- Family history of digestive issues
- Family history of autoimmunity
- Acquired antecedents
- Cigarette use from ages 12 to 31
- Multiple courses of antibiotics
- Clinical Findings
- I B D
- Hashimoto's thyroiditis
- GERD
- Recurrent otitis media
- Prenatal to birth
- Born via cesarean section
- Triggers and Precipitating Events
- Divorce
- Birth of first child
- High family conflict, absence of biological father
- Mediators and Perpetuators
- Chronic stress
- Ultra-processed food and three diet sodas per day
- Proton pump inhibitors

Functional Medicine Matrix

The Patient's Story

Antecedents (Inherited & Acquired)

- FHx of digestive issues & autoimmunity
- Antibiotic use, c-section

Triggers & Precipitating Events

- Childhood family conflict
- Divorce

Mediators/Perpetuators

- Chronic stress
- PPIs

Clinical Imbalances and Physiological Dysfunctions

Assimilation

- Dyspepsia
- GERD
- IBD

Structural Integrity

- Intestinal barrier dysfunction

Communication

- Thyroid dysfunction

Defense & Repair

- Autoimmune activity
- Hashimoto's
- IBD

Energy

- Fatigue

Mitigation & Clearance

- Cigarette history

Transport



Modifiable Lifestyle Factors

Sleep

- 5-6 hours a night, interrupted due to nighttime diarrhea

Physical Activity

Nutrition

- Skips meals, Ultra-processed foods, Diet sodas

Stressors

Relationships

- Divorce

Long Description: The Functional Medicine Matrix

- A completed Functional Medicine Matrix for Patient C
- The Patient's Story
- Antecedents
- Inherited and Acquired
 - Family history of digestive issues and autoimmunity
 - Born via Cesarean section
- Triggers and Precipitating Events
 - Childhood family conflict, Divorce
- Mediators and Perpetuators
 - Chronic stress, P P I's
- Modifiable Lifestyle Factors
- Sleep
 - 5-6 hours a night, interrupted due to nighttime diarrhea
- Nutrition
 - Skips meals, ultra-processed foods, diet sodas
- Relationships
 - Divorce
- Clinical Imbalances and Physiological Dysfunctions
- Defense and repair
 - Autoimmune activity, Hashimoto's, I B D
- Energy
 - Fatigue
- Mitigation and clearance
 - Cigarette history
- Communication
 - Thyroid dysfunction
- Structural integrity
 - Intestinal barrier dysfunction
- Assimilation
 - Dyspepsia, GERD, I B D
- M E S
 - Difficulty concentrating
 - Grief and anger

Recommended IFM Toolkit Items

Medical Symptoms Questionnaire (MSQ)

Patient Name _____

Instructions
Use this point scale to rate each of the following symptoms based on how often you experience them and how severe the effect is.

0 = Never or almost never have the symptom
1 = Occasionally have it; effect is not severe
2 = Occasionally have it; effect is severe

3
4

Head

Headaches
Faintness
Dizziness
Insomnia
Total

Eyes

Watery or itchy eyes
Swollen, reddened, or sticky eyelids
Bags or dark circles under eyes
Blurred or tunnel vision (doesn't include nearsightedness or farsightedness)
Total

Ears

Itchy ears
Earaches, ear infections
Drainage from ear
Ringing in ears, hearing loss
Total

Nose

Stuffy nose
Sinus problems
Hay fever
Sneezing attacks
Excessive mucus formation
Total

Adult Intake Questionnaire

General Information
Current Health Concerns
Allergies
History

Patient's Birth/Childhood History
Dental History
Environmental/Detoxification History
Obstetric History
Menstrual History
Sexual History

Family History
Medical History: Illnesses/Conditions
Medical History (continued)
Symptom Review
Medications/Supplements

Current medications (include prescription and over-the-counter)
Nutritional supplements (vitamins/minerals/herbs etc.)

Lifestyle Review
Sleep
Physical Activity
Nutrition
Tobacco
Alcohol
Other Substances
Stress
Relationships

Meaning and Purpose
Readiness Assessment and Health Goals
Readiness Assessment
Health Goals



PROMIS Questionnaires

The Patient-Reported Outcomes Measurement Information System (PROMIS) is a collection of validated questionnaires that can be used by clinicians and researchers to evaluate a patient's physical, mental, and social health in a standardized way. PROMIS questionnaires examine many components of a patient's health, including physical function, mental health, social health, support, cognition, life satisfaction, and more.

About PROMIS for Healthcare

- PROMIS questionnaires are available in multiple languages.
- Questionnaires can be used by practitioners and patients.
- Questionnaires provide practical functions for the general population and healthcare providers.

The PROMIS-Global Health (PGH) Questionnaire

- A widely available, quick, 10-question questionnaire.
- Focuses on function; specific quality of life making it comparable to other measures.
- Helps clinicians measure and address patient-reported outcomes through self-reported outcomes.
- Provides data to support functional medicine models.
- Provides a standardized measure for research studies to collect patient-reported outcomes.



Diet, Nutrition & Lifestyle Journal—3 Day

Patient Name _____
Date _____
Food Plan Type _____

Day 1

Day Event	Food & Drink Intake (include type, amount, brand)	Macronutrients (PFC) and Phytonutrients
Rising Time		P _____ F _____ C _____ OR O O O Y O G O B/P/BL O W/T/BR
Breakfast Time		P _____ F _____ C _____ OR O O O Y O G O B/P/BL O W/T/BR
Mid-AM Snack Time		P _____ F _____ C _____ OR O O O Y O G O B/P/BL O W/T/BR
Lunch Time		P _____ F _____ C _____ OR O O O Y O G O B/P/BL O W/T/BR
Mid-PM Snack Time		P _____ F _____ C _____ OR O O O Y O G O B/P/BL O W/T/BR
Dinner Time		P _____ F _____ C _____ OR O O O Y O G O B/P/BL O W/T/BR
PM Snack Time		P _____ F _____ C _____ OR O O O Y O G O B/P/BL O W/T/BR
Bed Time		P _____ F _____ C _____ OR O O O Y O G O B/P/BL O W/T/BR

P: Protein; F: Fat; C: Carbohydrates; R: Red; O: Orange; Y: Yellow; G: Green; B/P/BL: Blue/Purple/Black; W/T/BR: White/Tan/Brown

Sleep & Relaxation	Exercise & Movement	Stress	Relationships
Sleep Quantity _____ (hrs) Quality: ☐ Poor ☐ Fair ☐ Good	Type, Duration, & Intensity ☐ Aerobic: ☐ Strength: ☐ Flexibility:	Stress reduction practices: Stressors:	Supporting: Non-supporting:

Mental	Emotional	Spiritual

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Accessible versions in your toolkit.

Long Description: Recommended Toolkit Items

- Toolkit items:
- M S Q – Medical Symptoms Questionnaire
- Adult Intake Questionnaire
- P R O M I S Questionnaires
- Diet, Nutrition, and Lifestyle Journal –3 Day
- Accessibility versions available in toolkit.

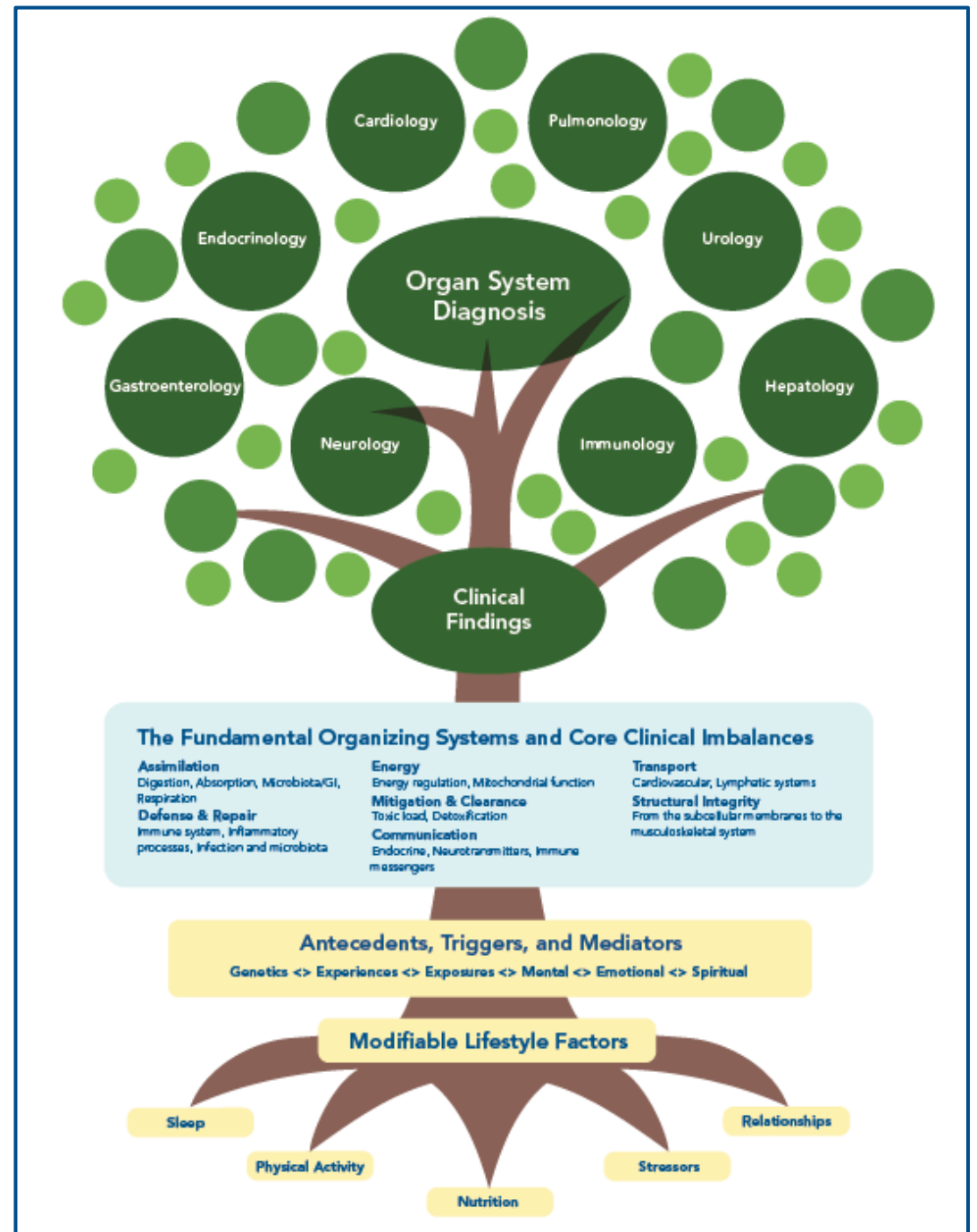
Clinical Focus

- Summary, conclusion, key takeaways, and clinical flow path.



Organizing the Data Nodes of the Matrix:

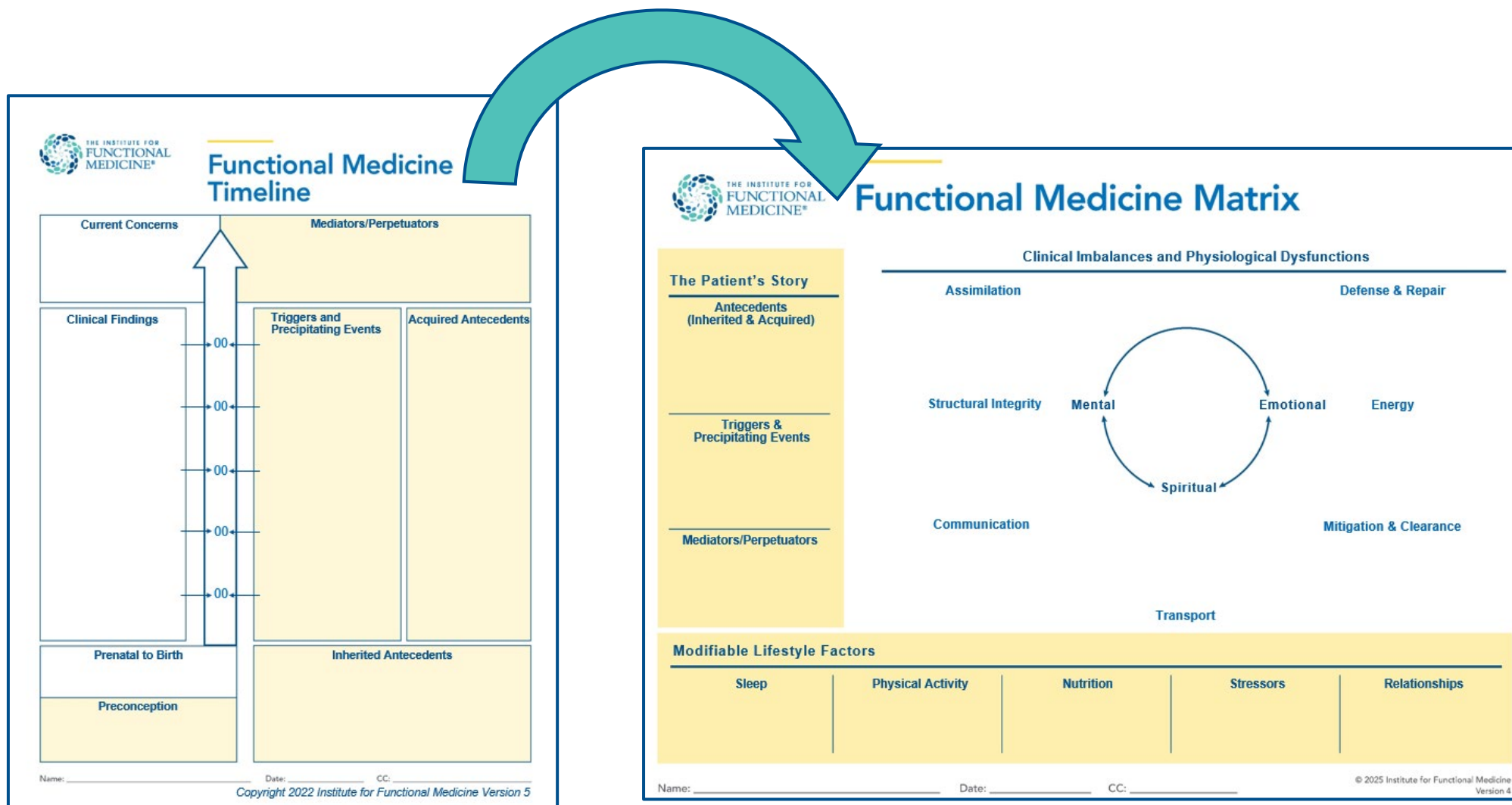
- Distinguishing features that inform the Functional Medicine Patient Evaluation



Long Description

- The practitioner can organize the distinguishing features of the functional medicine patient evaluation like the structure of a tree. The roots of the tree represent the modifiable lifestyle factors including sleep, physical activity, nutrition, stressors, and relationships. Moving upward, the base of the trunk represents antecedents, triggers, and mediators including genetics, epigenetics, exposures, mental health, emotional health, and spiritual health. The middle of the trunk is the fundamental organizing systems and core clinical imbalances. These are: assimilation, which includes digestion, absorption, microbiota or G I, and respiration; defense and repair, which includes the immune system, inflammatory processes, and infection and microbiota; energy, which includes energy regulation and mitochondrial function; mitigation and clearance, which includes toxic load and detoxification; communication, which includes endocrine, neurotransmitters, and immune messengers; transport, which includes cardiovascular and lymphatic systems; and structural integrity from the subcellular membranes to the musculoskeletal system. The top of the trunk represents clinical findings, and the top branches are the organ system diagnosis. The leaves represent the organ systems including neurology, gastroenterology, endocrinology, cardiology, pulmonology, urology, hepatology, and immunology.

Transferring Clinical Findings, ATMs, and MLFs from the Timeline to the Matrix

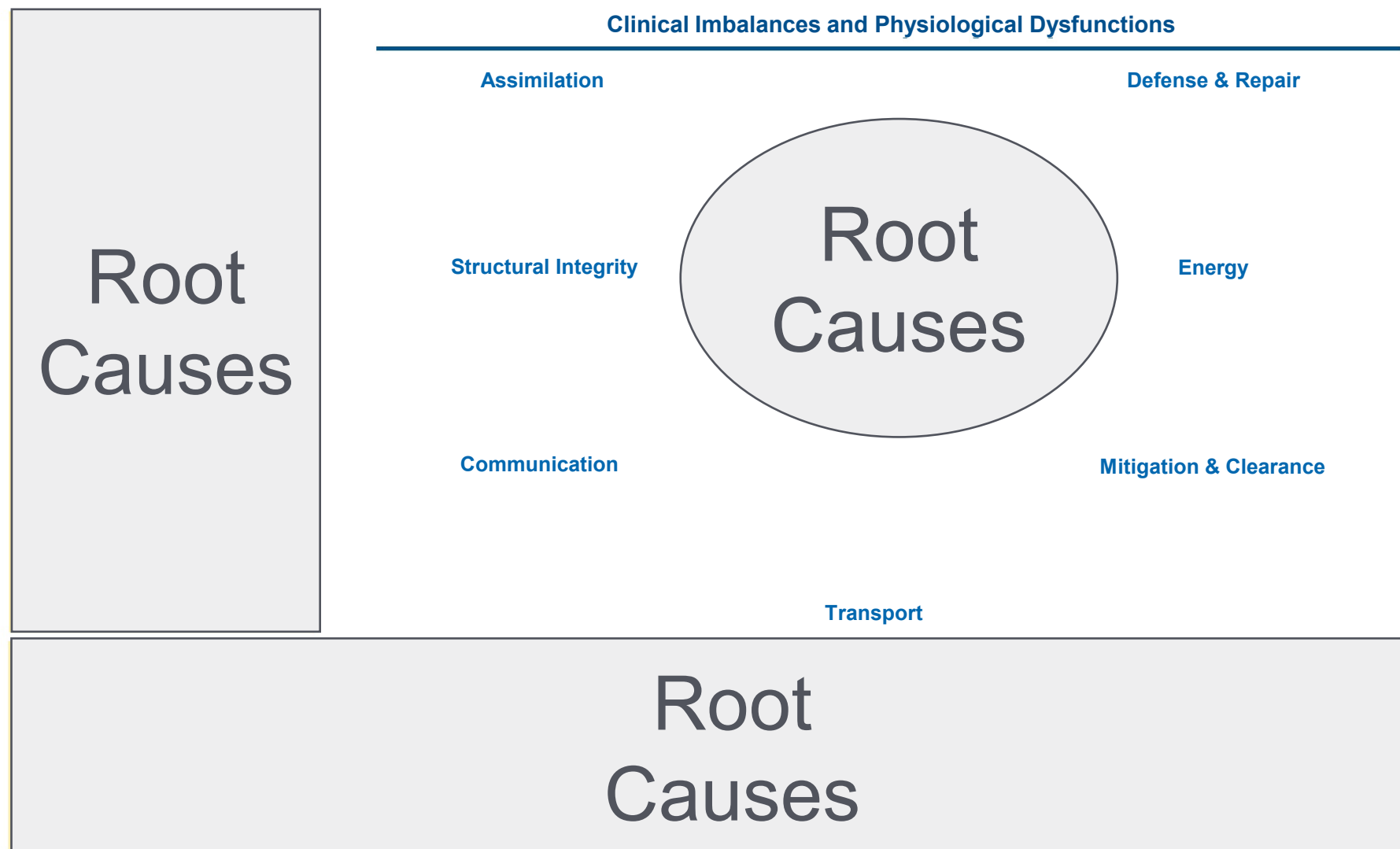


Long Description: Transferring Clinical Findings, ATMs, and MLFs from the Timeline to the Matrix

- An arrow pointing from a blank Functional Medicine Timeline to a blank Functional Medicine Matrix.



Functional Medicine Matrix



Name: _____ Date: _____ CC: _____

Long Description: The Functional Medicine Matrix

- A blank Functional Medicine Matrix with the phrase Root Causes over each section of the Matrix. Organizing the patient's clinical information in the Matrix allows the clinician to identify these underlying root causes.



Functional Medicine Matrix

The Patient's Story

Antecedents
(Inherited & Acquired)

Triggers &
Precipitating Events

Mediators/Perpetuators

Effects
(Clinical Findings)

Modifiable Lifestyle Factors

Sleep

Physical Activity

Nutrition

Stressors

Relationships

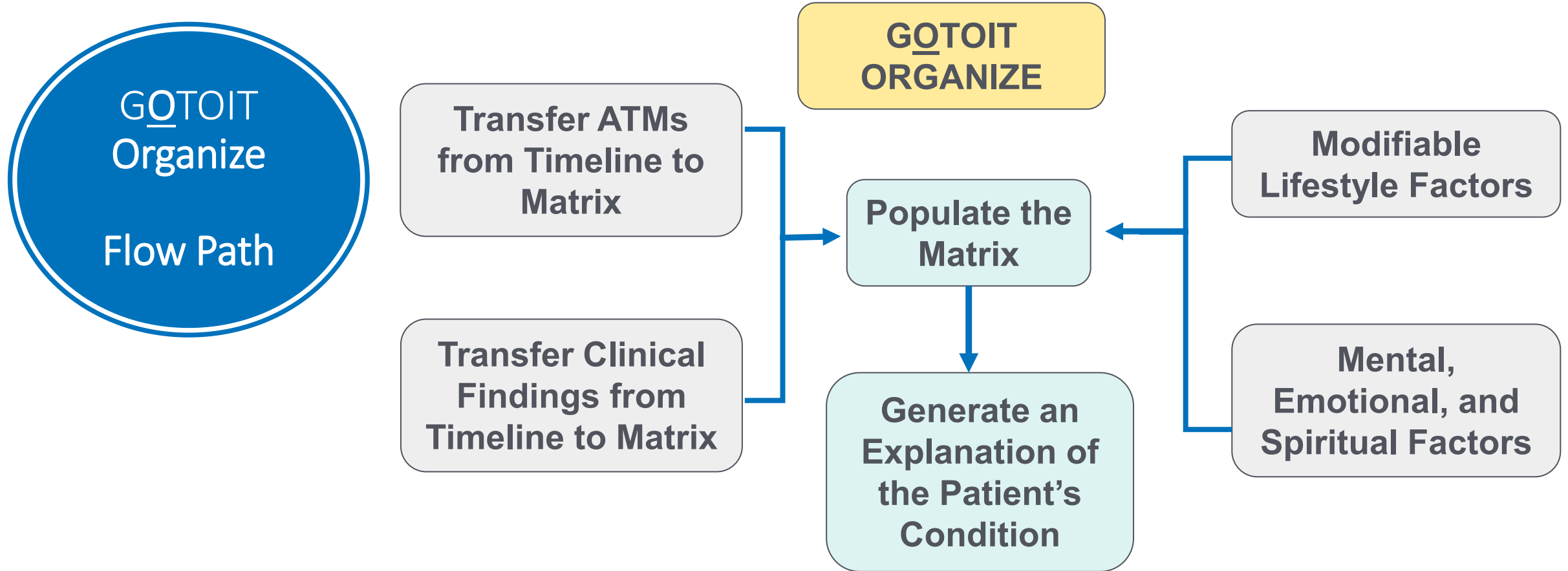
Name: _____ Date: _____ CC: _____

Long Description: The Functional Medicine Matrix

- A blank Functional Medicine Matrix with the phrase Effects Clinical Findings over the Clinical Imbalances and Physiological Dysfunctions.

Key Clinical Takeaway:

Flow Path for the Organize Step of GOTOIT



Long Description: Flow Path for the Organize Step of GOTOIT

- Perform the following to populate the Matrix:
- Transfer A T M's from Timeline to Matrix
- Transfer Clinical Findings from Timeline to Matrix
- Add Modifiable Lifestyle Factors and Mental, Emotional, and Spiritual Factors to the Matrix
- Once the Matrix is populated, generate an explanation of the patient's condition

The Functional Medicine Clinical Method

WHY IT IS CRITICAL TO “ORGANIZE” THE DYSFUNCTIONS ON THE MATRIX

GOTOIT Functional Medicine Framework



Long Description: The Functional Medicine Clinical Method

- The Go to it functional medicine framework
- G
- Gather oneself and clinical findings on the I F M functional medicine timeline
- O
- Organize clinical findings and imbalances on the I F M functional medicine matrix
- T
- Tell the patient's story to ensure a shared understanding
- O
- Order and prioritize interventions feasible for the patient
- I
- Initiate referrals, evaluative tests, and therapeutic modalities
- T
- Track and adjust evaluate and therapeutic regimens
- Two arrows pointing from Organize to Tell and Order highlight how organizing the patient's data on the Matrix prepares the clinician to retell the patient's story and prioritize interventions.

ORGANIZE

The Functional Medicine Matrix

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE